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Spatial charting of single-cell transcriptomes in tissues

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Single-cell RNA sequencing methods can profile the transcriptomes of single cells but cannot preserve spatial information. Conversely, spatial transcriptomics assays can profile spatial regions in tissue sections, but do not have single-cell resolution. Here, we developed a computational method called CellTrek that combines these two datasets to achieve single-cell spatial mapping through coembedding and metric learning approaches. We benchmarked CellTrek using simulation and in situ hybridization datasets, which demonstrated its accuracy and robustness. We then applied CellTrek to existing mouse brain and kidney datasets and showed that CellTrek can detect topological patterns of different cell types and cell states. We performed single-cell RNA sequencing and spatial transcriptomics experiments on two ductal carcinoma in situ tissues and applied CellTrek to identify tumor subclones that were restricted to different ducts, and specific T cell states adjacent to the tumor areas. Our data show that CellTrek can accurately map single cells in diverse tissue types to resolve their spatial organization.

Single-cell RNA sequencing (scRNA-seq) methods have greatly expanded our understanding of the gene expression programs of diverse cell types and their role in development and disease[1–5]. However, scRNA-seq loses cellular spatial information during the tissue dissociation step, which is critical for understanding cellular microenvironment and cell–cell interactions[6–8]. While spatial sequencing methods, including spatial transcriptomics (ST)[9] and Slide-seq[10], can profile gene expression spatially across tissue sections, they are limited to measuring small regions with mixtures of cells and cannot easily provide single-cell information. To address this issue, computational approaches (for example, cell2location, RCTD (robust cell type decomposition)) have been designed to deconvolute ST spots into proportions of different cell types[6,11–16]. However, spatial deconvolution methods are limited to inferring only cell type proportions for each spot, and cannot achieve single-cell resolution. Additionally, deconvolution methods have limited ability to further resolve cell types into more granular 'cell states' (expression programs) that reflect different biological functions. Finally, most deconvolution methods can only predict categorical labels and cannot infer continuous cell information (for example, lineage trajectories, gene signatures) at a spatial resolution.

Here, we introduce CellTrek—a computational toolkit that can directly map single cells back to their spatial coordinates in tissue sections based on scRNA-seq and ST data. This method is distinct from ST deconvolution, enabling a more flexible and direct investigation of single-cell data with spatial topography. The CellTrek toolkit also provides two downstream analysis modules, including SColoc for spatial colocalization analysis and SCoexp for spatial coexpression analysis. We benchmarked CellTrek using simulations and in situ datasets. We then applied CellTrek to existing datasets from normal mouse brain[17] and kidney[18] tissues as well as to data that we generated from two human ductal carcinoma in situ (DCIS) samples to study the spatial organization of cell types/states at a single-cell resolution.

Results

Overview of CellTrek toolkit. CellTrek first integrates and coembeds ST and scRNA-seq data into a shared feature space (Fig. 1; Methods). Using the ST data, CellTrek trains a multivariate random forests (RF) model[19] to predict the spatial coordinates using shared dimension reduction features. A spatial nonlinear interpolation on ST data is introduced to augment the spatial resolution. The trained model is then applied to the coembedded data to derive an RF distance matrix that measures the expression similarities between ST spots and single cells supervised by spatial coordinates. Based on the RF distance matrix, CellTrek produces a sparse spot-cell graph using mutual nearest neighbors (MNN) after thresholding. Finally, CellTrek transfers spatial coordinates for cells from their neighbor spots. To improve the compatibility, CellTrek can accept any cell-location probability/distance matrix calculated from other methods (for example, novoSpaRc[20]) as an input for cell spatial charting. Additionally, we provide a graphical user interface (GUI) for interactive visualization of the resulting CellTrek map.

To recapitulate spatial relationships between different cell types, we developed a downstream computational module, SColoc, which summarizes the CellTrek result into a graph abstraction (Supplementary Fig. 1a; Methods). Three approaches, Kullback-Leibler divergence (KL), Delaunay triangulation (DT) and K-nearest neighbor distance (KD), are provided to calculate spatial dissimilarity between cell types. Based on the dissimilarity matrix, SColoc can construct a minimum spanning tree (MST) that represents a simplified spatial cellular proximity. The above steps will be executed iteratively on bootstrapping samples to generate consensus matrices. Thereafter, a graph will be rendered through a GUI with tunable edge thresholding and color mapping functions. Additionally, SColoc provides a K-distance metric for measuring the spatial distance of cells to a selected reference group.

To investigate whether different expression programs are distributed across different topographic areas, we developed SCoexp,

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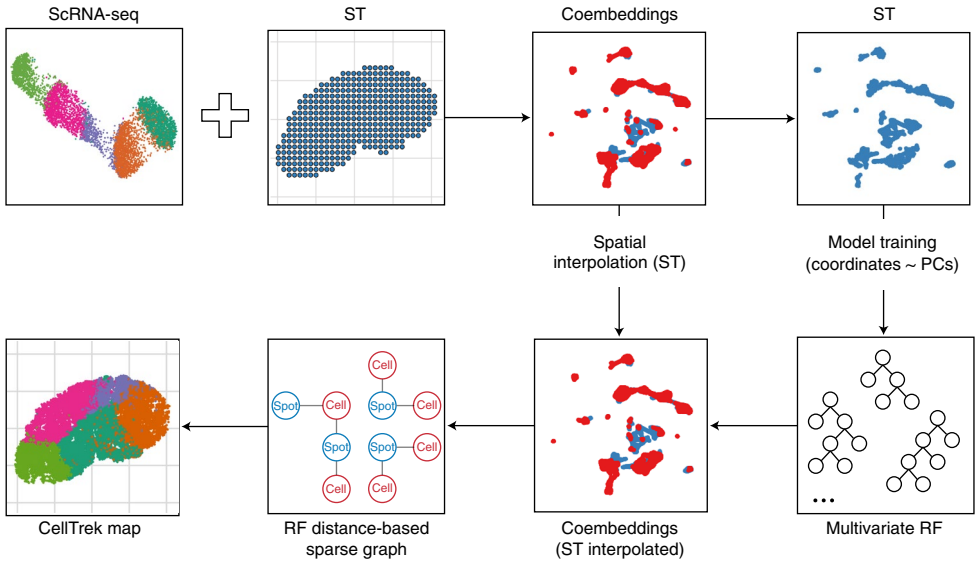

Fig. 1 | Overview of the CellTrek workflow. CellTrek first coembeds scRNA-seq and ST datasets into a shared latent space. Using the ST data, CellTrek trains a multivariate RF model with spatial coordinates as the outcome and latent features as the predictors. A 2D spatial interpolation on the ST data is introduced to augment the ST spots. The trained RF model is then applied to the coembedded data (ST interpolated) to derive an RF distance matrix, which will be converted into a sparse graph using MNN. Based on the sparse graph, CellTrek transfers the coordinates to single cells from their neighboring ST spots.

which uses the CellTrek coordinates to detect coexpression gene modules within the cells of interest (Supplementary Fig. 1b; Methods). First, SCoexp calculates a spatial kernel weight matrix. Using this weight matrix, SCoexp calculates spatial-weighted gene coexpression matrix. Thereafter, SCoexp utilizes consensus clustering[21] (CC) or weighted correlation network analysis[22] (WGCNA) to identify gene modules. For the identified modules, we can calculate module scores and investigate their spatial organizations.

Benchmarking and simulations. To benchmark the performance of CellTrek, we exploited three spatial datasets: (1) a simulated scRNA-seq dataset with customized spatial patterns (Supplementary Fig. 2a,b), (2) a fluorescence in situ hybridization (FISH)-based single-cell dataset of the *Drosophila* embryo[20] (Supplementary Fig. 2d,e) and (3) a seqFISH dataset of mouse embryo[23] (Supplementary Fig. 2g,h). We generated three corresponding ST datasets with each spot aggregating the five spatially nearest cells (Supplementary Fig. 2c,f,i).

We applied CellTrek to the scRNA-seq and ST data to reconstruct their spatial cellular maps. We then compared CellTrek with two additional cell charting methods: (1) NVSP-CellTrek, which uses a reference-based novoSpaRc[20]—a spatial reconstruction method—to calculate a cell spatial probability matrix, which then uses CellTrek to produce a spatial map; and (2) Seurat[24] coordinate transfer (SrtCT), which uses the data transfer approach to transfer ST coordinates to single cells. Both CellTrek and NVSP-CellTrek reconstructed the original spatial pattern of the simulated data, while SrtCT reconstructed only a rough spatial cellular relationship (Supplementary Fig. 3a). Compared with NVSP-CellTrek, CellTrek mapped more cells with higher spatial density. To quantitatively evaluate these methods, we compared the spatial density of the cell charting results with the original spatial distribution across different cell types using the KL-divergence. Both CellTrek and NVSP-CellTrek achieved good performance with low KL-divergences, while SrtCT showed much higher discrepancies to the reference distribution (Supplementary Fig. 3b). In the *Drosophila* embryo data, CellTrek accurately reconstructed the original spatial layout with the lowest KL-divergences among three approaches (Supplementary Fig. 3c,d).

We further investigated several known *Drosophila* embryogenic genes in the CellTrek results and found spatial patterns consistent with those of the previous study[20] (Supplementary Fig. 3e). In the mouse embryo data, we found that CellTrek and NVSP-CellTrek accurately reconstructed the original spatial structure, while CellTrek showed slightly higher KL-divergences in groups 5, 9 and 17 (Supplementary Fig. 3f,g). To investigate whether CellTrek could reveal the developing spatial patterns of the mouse embryo, we selected a group of gut tube cells and found spatial expression patterns in some marker genes[23] (Supplementary Fig. 3h). We then performed a trajectory analysis using Monocle2 (refs. [25,26]), which showed that the pseudotime reflected the spatial developing pattern of the gut tube cells along with the anterior–posterior axis[23] (Supplementary Fig. 3i).

We next assessed the performance of CellTrek on simulated data under three different simulation settings: (1) read counts, (2) spatial randomness and (3) tissue densities. We evaluated CellTrek performance using KL-divergence and Pearson's correlation on the cell spatial coordinates between the CellTrek map and the reference. Across different simulation conditions, CellTrek achieved good spatial reconstruction performances (Supplementary Fig. 4a,c,e,g,i) and showed lower KL-divergences and higher correlations compared with the permutation distributions (Supplementary Fig. 4b,f,j). However, increasing the spatial randomness will affect the performance of CellTrek and decrease the statistical significance (Supplementary Fig. 4f), whereas decreasing the read counts or the spot/cell density will result in sparse cellular maps (Supplementary Fig. 4a,b,g–j). Overall, these data suggest that CellTrek is a robust method for single-cell spatial mapping under different experimental conditions.

Topological organizations of mouse brain cells. We applied CellTrek to public mouse brain scRNA-seq (Smart-seq2)[17] and ST datasets (Visium, 10x Genomics). We compared CellTrek with NVSP-CellTrek and SrtCT approaches. CellTrek reconstructed a clear layer structure of laminar excitatory neuron subtypes, in the order of L2/3 intratelencephalic (IT), L4, L5 IT, L6 IT, L6 corticothalamic (CT) and L6b, which matched to the cerebral cortex structure

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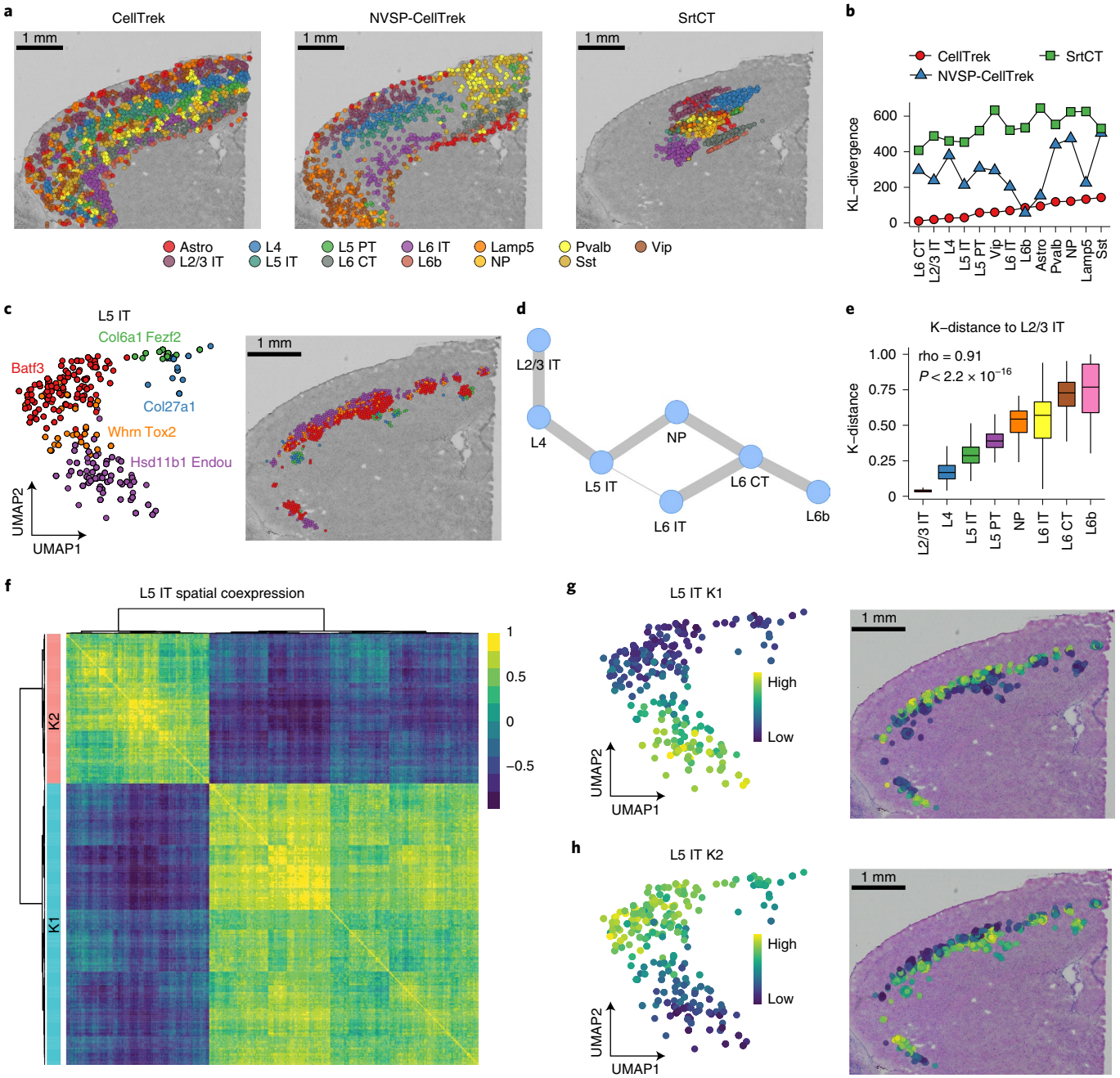

Fig. 2 | CellTrek reconstructs spatial organization in a mouse brain tissue. a, Comparison of CellTrek, NVSP-CellTrek and SrtCT results for single-cell spatial charting in a mouse brain tissue. **b**, KL-divergence of spatial cell charting methods for each cell type using SrtLT as a reference. **c**, UMAP (left) and CellTrek map (right) of scRNA-seq data of L5 IT cell states. **d**, Spatial colocalization graph of glutamatergic neurons using SColoc. **e**, CellTrek-based spatial K-distance of glutamatergic neurons to L2/3 IT cells (L2/3 IT $n=739$; L4 $n=576$; L5 IT $n=690$; L5 PT $n=365$; NP $n=182$; L6 IT $n=695$; L6 CT $n=721$; L6b $n=153$). Boxplots show the median with interquartile ranges (25–75%); whiskers extend to 1.5× the interquartile range from the box. We performed a Spearman correlation test. **f**, Spatial coexpression modules (K1 and K2) identified in L5 IT cells using SCoexp. **g**–**h**, UMAPs of L5 IT cells showing the K1 module activity scores (**g**) and the K2 module activity scores (**h**) and their corresponding CellTrek maps.

(Fig. 2a and Supplementary Data 1). NVSP-CellTrek showed a similar spatial layer trend, thus demonstrating the flexibility and consistency of the CellTrek approach (Fig. 2a). However, NVSP-CellTrek resulted in a sparse cell mapping in some areas. SrtCT failed to accurately project cell locations to the histological image (Fig. 2a). We then employed Seurat label transfer (SrtLT) to predict the spatial distribution of each cell type as our reference[27]. KL-divergences between cell charting results and the reference suggested that CellTrek successfully recovered spatial cellular structures with the lowest KL-divergences overall (Fig. 2b).

Next, we asked whether CellTrek could further uncover topological patterns of cell states within the same cell type. For example, L5 IT cells contain five expression states and showed a continuous trend on the uniform manifold approximation and projection (UMAP) in the order of Hsd11b1-Endou, Whrn-Tox2, Batf3, Col6a1-Fezf2 and Col27a1 (Fig. 2c, left). The L5 IT CellTrek map

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discovered a refined sublayer architecture (Fig. 2c, right), which is consistent with the results of a previous study[28]. To summarize the cell spatial colocalizations, we applied the SColoc to the CellTrek result using KL-based MST consensus graph. Glutamatergic neuron cell types formed a linear backbone of the graph in the order of the layer structures (Fig. 2d). Spatial K-distance to the L2/3 IT cells showed an increasing trend in the same order of the graph (Spearman's rho = 0.91, $P < 2.2 \times 10^{-16}$) (Fig. 2e).

Using SCoexp, we then investigated how genes were coexpressed spatially within L5 IT cells. Two coexpression modules (K1, K2) were identified and enriched in different biological functions (Fig. 2f and Supplementary Fig. 5a,b). The K1 module was highly active in cell states Hsd11b1-Endou, Whrn-Tox2 and spatially located in the outer layer, while the K2 module was highly active in Col27a1, Col6a1-Fezf2 and Batf3 (Fig. 2c,g,h) and located mainly in the inner layer (Fig. 2g,h). These results show that SCoexp is able to identify subtle transcriptional differences within the same cell type and infer their topological heterogeneity.

Spatial cell charting of the mouse hippocampus. We also applied CellTrek to Slide-seq v.2 (ref. [29]) and scRNA-seq data[30] from the mouse hippocampus. Unsupervised clustering of the Slide-seq data identified 12 clusters (G01–G12) with a highly organized spatial structure (Supplementary Fig. 6a). CellTrek mapped single cells to their spatial locations (Supplementary Fig. 6b), which is consistent with the Slide-seq clusters. Notably, G06 matched with the Cornu Ammonis (CA) areas (Supplementary Fig. 6c), while CellTrek revealed a sequential mapping of the principal cells CA1, CA2 and CA3 that were not resolved by the Slide-seq clustering alone (Supplementary Fig. 6d). These results show that CellTrek can be applied broadly to different spatial genomic platforms to achieve more refined spatial cellular resolution.

Spatial reconstruction of a mouse kidney tissue. We applied CellTrek to a public mouse kidney data[18] and compared it with NVSP-CellTrek and SrtCT. CellTrek accurately reconstructed cellular spatial structures with distinct cell types located in different histological zones (for example, cortex, outer medulla and inner medulla) (Fig. 3a and Supplementary Data 2). NVSP-CellTrek showed similar spatial patterns compared with CellTrek, whereas SrtCT could not reconstruct accurate spatial organizations of the mouse kidney cells (Fig. 3a). Using SrtLT as a reference, both CellTrek and NVSP-CellTrek achieved overall low KL-divergence compared with SrtCT. NVSP-CellTrek showed higher KL-divergence for VSMC and RenaCorp cells (Fig. 3b). To further study the spatial expression dynamics, we inferred the trajectories of both ProxTub and DistTub cells and mapped their pseudotime spatially using CellTrek. For ProxTub cells, we observed a continuous spatial trajectory that started from the outer part of the cortex to the inner part (Fig. 3c). This continuous anatomic change of ProxTub cells is consistent with previous studies[31,32]. Similarly, DistTub cells also showed a continuous trajectory with a clear spatial pattern (Fig. 3d). Collectively, these results show that CellTrek can resolve the topological arrangements of continuous expression programs of single cells in tissues.

We next summarized a cell spatial graph using SColoc. ProxTub cells were identified as the hub and connected to the RenaCorp, DistTub and other cell types (Fig. 3e). The consensus heatmap and hierarchical clustering showed similar patterns to the graph abstraction (Fig. 3f). Since the scRNA-seq data were collected from different zonal microdissections of the mouse kidneys[18], we asked if CellTrek could recapitulate the experimental zonal information without previous knowledge. Based on the CellTrek result, we calculated the K-distance for TLLH, DistTub and Prin cells to a group of cells from the center region. A consistent trend was observed that K-distances decreased from cortex to outer medulla then to inner medulla, suggesting that CellTrek successfully revealed the zonal structure of the mouse kidney (Fig. 3g). Further, in the DistTub cells, we identified two distinct spatial coexpression modules (K1 and K2) using SCoexp (Fig. 3h). The K1 module was enriched in metabolic pathways, renal system development and highly correlated with some distal convoluted tubule (DCT) genes (for example, *Wnk1* and *Slc12a3*)[33,34] (Supplementary Fig. 5c). In contrast, K2 was enriched in cell-matrix pathways, purine metabolic pathways and correlated with distal straight tubule (DST) canonical genes (for example, *Slc12a1* and *Umod*)[34] (Supplementary Fig. 5d). These two modules displayed distinct patterns on the CellTrek map. K1 was highly active in the cortex area, whereas K2 was active in the medulla, which is consistent with the anatomic localizations of DCT and DST (Fig. 3i,j).

We further asked whether CellTrek could improve our understanding of cell–cell communications by leveraging the spatial information. We conducted a cell–cell interaction analysis on the scRNA-seq data using CellChat[35] and used the SColoc graph (Fig. 3e) to filter cell–cell pairs, assuming that colocalized cells have higher chances to interact with each other. Compared with the raw CellChat results, which predicted many nonspecific interactions with all cell types interacting with each other (Supplementary Fig. 5e), the spatial filtering provided more concise and specific cell–cell interactions (Supplementary Fig. 5f). We identified several known interactions, including *Vegfa*, which is expressed by the ProxTub interacted with its receptors *Flt1* and *Kdr*, which are expressed by Vasc (Supplementary Fig. 5g)[36–39].

Spatial subclone heterogeneity in a DCIS breast cancer. We applied 3′ scRNA-seq (10x Genomics) and ST (Visium, 10x Genomics) to a DCIS sample (DCIS1) to profile 6,828 single cells and 1,567 ST spots. For the scRNA-seq data, clustering and differential expression (DE) analyses identified five main cell types, including epithelial, endothelial, fibroblast, myeloid and natural killer (NK)/T cells (Supplementary Fig. 7a). We applied CopyKAT[40] to infer copy number profiles from the scRNA-seq data. We observed some clonal copy number alterations (CNAs) across all tumor cells, including gains on chromosomes 3q (*PIK3CA*), 8q (*MYC*) and 19p (*STK11*) and losses on chromosomes 8p (*PPP2R2A*), 10q (*PTEN*) and 14q (*AKT1*) (Fig. 4a). UMAP and dbscan clustering of the CNA profiles

Fig. 3 | CellTrek reconstructs spatial organization in a mouse kidney tissue. a, Comparison of CellTrek, NVSP-CellTrek and SrtCT results for single-cell spatial charting in a mouse kidney tissue. DistTub, distal tubule cells; T, T cells; ProxTub, proximal tubule cells; VSMC, vascular smooth muscle cells; Inter, intercalated cells; Prin, principal cells; TLLH, the loop of Henle; Vasc, vascular cells; Macro, macrophages; RenaCorp, renal corpuscle cells. **b**, KL-divergence of spatial cell charting methods for each cell type using SrtLT as a reference. **c**, Trajectory analysis for proximal tubule cells (left) and spatial mapping of the pseudotime values in the tissue section (right). **d**, Trajectory analysis for distal tubule cells (left) and spatial mapping of the pseudotime values in the tissue section (right). **e**, Spatial colocalization graph of different renal cell types using SColoc. **f**, Spatial consensus matrix of different renal cell types. **g**, CellTrek-based spatial K-distance of TLLH, DistTub and Prin cells to the tissue center cells across experimental zonal dissections (left). Center cells as reference are shown on the right panel. Kruskal–Wallis rank sum tests were performed (cortex $n = 1{,}169$; outer medulla $n = 1{,}423$; inner medulla $n = 1{,}661$). Boxplots show the median with interquartile ranges (25–75%); whiskers extend to 1.5× the interquartile range from the box. **h**, Spatial coexpression modules (K1 and K2) identified in distal tubule cells using SCoexp. **i**–**j**, UMAPs of distal tubule cells showing the K1 module activity scores (**i**) and the K2 module activity scores (**j**) and their corresponding CellTrek maps.

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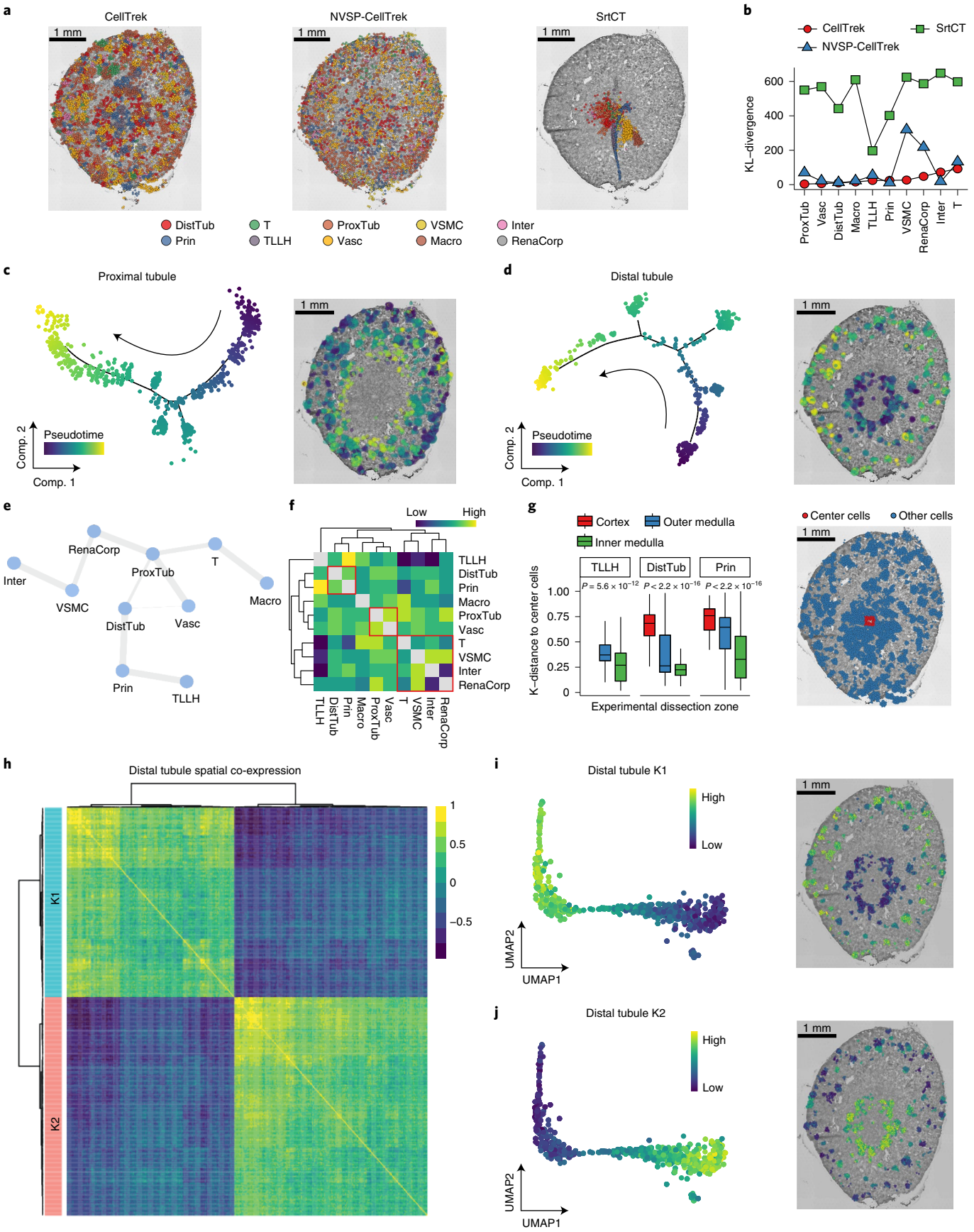

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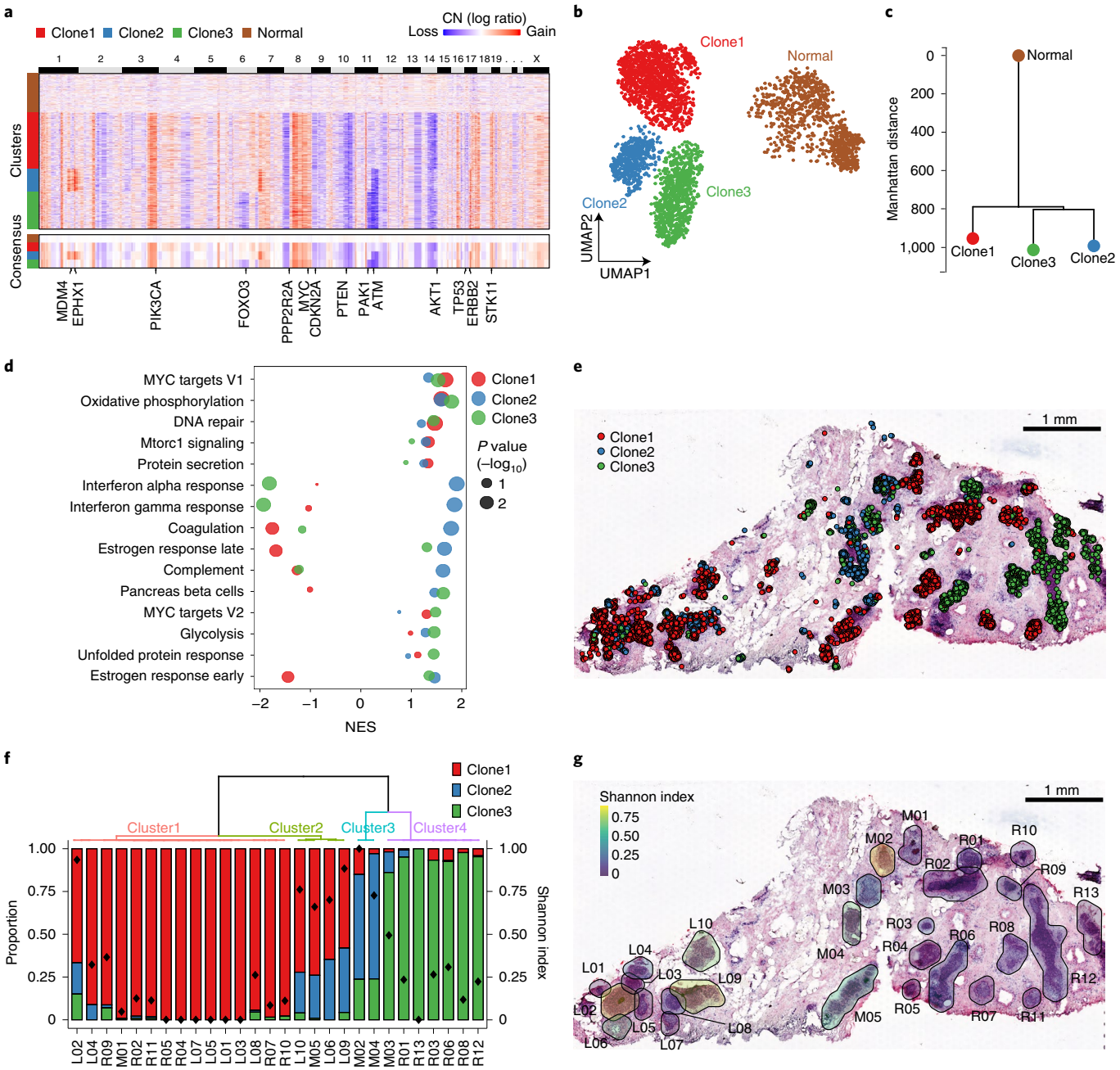

Fig. 4 | CellTrek identifies the spatial subclone heterogeneity in DCIS1. a, A heatmap of copy number (CN) profiles inferred by CopyKAT on the scRNA-seq data in DCIS1. The lower part represents a consensus CN profile of each cluster with some breast cancer-related genes annotated. **b**, CN-based UMAP of DCIS1. **c**, Phylogenetic tree based on the consensus CN profiles. **d**, Hallmark GSEA analysis of the expression data from three tumor subclones. *P* values were determined by an adaptive multi-level split Monte-Carlo method and then adjusted by the Benjamini-Hochberg method. **e**, Spatial cell charting of three tumor subclones using CellTrek. **f**, Tumor subclonal compositions in different ducts. The diamond symbol in each bar represents the Shannon index which measures the diversity of tumor subclones. **g**, H&E image of the DCIS tissue section with Shannon diversity index for each duct.

identified three main tumor subclones (clone1–3) with some distinct alterations, including 17q (*ERBB2*) gain and 11q (*ATM*) loss in clone2 and clone3, 1q (*MDM4* and *EPHX1*) gain in clone2 and 6q (*FOXO3*) loss in clone3 (Fig. 4a,b). Based on the consensus CNA profiles, we constructed a phylogenetic tree, which showed that clone1 was an earlier subclone that diverged from the main lineage, followed by clone2 and clone3 (Fig. 4c). Notably, these three subclones displayed transcriptional heterogeneity (Supplementary Fig. 7b). Hallmark gene set enrichment analysis[41] identified several common pathways across all three subclones, including MYC targets, oxidative phosphorylation and DNA repair (Fig. 4d). We also identified subclonal-specific signatures, including estrogen response pathways enriched in clone2 and clone3, and interferon alpha/gamma response, coagulation and complement pathways enriched in clone2.

To understand the spatial distribution of three tumor subclones, we applied CellTrek to the scRNA-seq and ST data. Most tumor cells mapped to the DCIS regions on the hematoxylin and eosin (H&E) slide (Fig. 4e,g and Supplementary Data 3). Moreover, different tumor subclones mapped to different ductal regions, reflecting

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extensive spatial intratumor heterogeneity[42]. Specifically, clone2 was localized mostly to the middle (M) ducts, while clone3 was located primarily on the right (R) ducts and clone1 was spread across many ductal regions (Fig. 4e,g). Unsupervised clustering of the ST tumor spots identified five ST clusters that showed spatial and gene expression concordance to the tumor CellTrek map (Supplementary Fig. 7c–e). Based on the subclonal compositions of each duct, we performed a clustering analysis and calculated the Shannon index, resulting in four main ductal clusters with different subclone compositions and spatial patterns (Fig. 4f). Overall, ducts from the right part of the tissue displayed low clonal diversities, while some ducts from the middle and left regions showed higher clonal diversities (Fig. 4g).

We further investigated the spatial coexpression patterns of the tumor cells using SCoexp and identified three gene modules (K1, K2 and K3). The K1 module was high in clone1 and enriched in actin-related pathways (Supplementary Fig. 8a,e,f). CellTrek displayed that cells with high K1 scores corresponded to tumor clone1 spatially (Supplementary Fig. 8g). By contrast, K2 was high in clone2 and clone3, and was enriched in response to estradiol, mammary gland duct morphogenesis and some catabolic processes (Supplementary Fig. 8h–j). The K3 module was highly active in proliferating tumor cells and associated with cell cycle related processes (Supplementary Fig. 8b–d,k,l). The spatial mapping of the K3 score showed that proliferating tumor cells were located mostly near the peripheral regions of several ducts (Supplementary Fig. 8m). Taken together, these data show that the CellTrek toolkit can delineate topological maps of different tumor subclones and their expression programs in a DCIS tissue.

Spatial tumor-immune microenvironment of a DCIS tissue. In another DCIS sample with synchronous invasive components (DCIS2), we profiled 3,748 single cells and 2,063 ST spots. Unsupervised clustering and DE analyses identified ten clusters, including three epithelial clusters, endothelial, pericytes, fibroblasts, myeloid, NK/T, B and plasmacytoid dendritic cells (pDC) (Supplementary Fig. 9a,b). CopyKAT revealed an aneuploid epithelial cluster with CNAs (epithelial3) (Supplementary Fig. 9c). Histopathological analysis of the H&E image identified 11 ductal regions with tumor cells (T1–T11) and intervening areas that contained stromal and immune cells (Fig. 5a). To study the tumor-immune microenvironment, we focused on aneuploid cells and immune cells from the scRNA-seq data (Fig. 5b). Using CellTrek, we mapped most of the aneuploid cells to the histologically defined DCIS regions and immune cells to areas surrounding ducts and stromal regions (Fig. 5c and Supplementary Data 4). We found that some immune cells, including T, B, myeloid cells and pDC, were aggregated in the areas directly outside of the ducts, especially T1, T2, T6 and T7. Combining the CellTrek result with the H&E image, we posited the existence of tertiary lymphoid structures (TLS) in these regions. To further investigate this question, we calculated ST spot-level TLS scores[43,44] and found that spots with high TLS scores often corresponded to the mixed immune cell aggregations in our CellTrek map (Fig. 5c,d). Furthermore, we found that the ST-level TLS scores correlated positively with the charted immune cell counts (Spearman's rho = 0.36, $P = 1.2 \times 10^{-10}$) (Fig. 5e). Together, these results show that CellTrek is capable of reconstructing the spatial tumor-immune microenvironment based on the scRNA-seq and ST data.

Next, we found that some T cells were proximal to the tumor regions while some were distal. We further clustered T cells into six cell states, including the Naive T (NaiveT), CD4$^+$ T (CD4T), CD8$^+$ T (CD8T), regulatory T cells (Treg), exhausted CD4$^+$ T (CD4Te) and exhausted CD8$^+$ T (CD8Te) (Fig. 5g and Supplementary Fig. 9d). We investigated the distribution of these T cell states in the CellTrek map. Notably, the Tregs, CD4Te and CD8Te cells were mostly proximal to the tumor cells (Fig. 5f). We further constructed a spatial graph in the T cells and found that cells from the same lineages tended to colocalize spatially (Fig. 5h). We calculated T exhaustion scores and found that T cells with high exhaustion scores tended to localize near the tumor areas (Fig. 5i). K-distances of the T cells to their 15 nearest tumor cells showed an opposite trend to the T exhaustion scores on the UMAP (Fig. 5j,k). As expected, the immunosuppressive T cells (Treg, CD4Te and CD8Te) had higher exhaustion scores compared with the nonsuppressive T cells (Fig. 5l). We binarized T cells to tumor distal (TD) and tumor proximal (TP) groups based on their K-distances and found that the TP group showed significantly higher exhaustion scores than the TD group ($P = 1.1 \times 10^{-4}$, Fig. 5m), suggesting the presence of immunosuppressive microenvironment near the DCIS ductal regions. We also found a similar trend in which TP had higher exhaustion scores compared with TD for the CD4T and Treg cells, and an opposite trend for the NaiveT cells (Fig. 5n). The TD groups contained only few immunosuppressive T cells, which is consistent with our finding that exhausted T cells tend to colocalize near DCIS regions (Fig. 5n).

Reclustering of the myeloid cells identified four cell states, including conventional dendritic cells (cDCs), monocytes and two macrophage subpopulations (Macro1 and Macro2; Supplementary Figs. 9e and 10a). CellTrek projected most of the cDCs to the tumor proximal areas (Supplementary Fig. 10a). The spatial graph showed that the Macro2 cells were colocalized with Macro1 and cDC (Supplementary Fig. 10b). We calculated the K-distance of myeloid cells to the tumor cells (Supplementary Fig. 10c) and found that the cDCs displayed the lowest K-distances overall, while the Macro1 cells had higher K-distances. The K-distance density plot showed a similar trend (Supplementary Fig. 10d). We further examined the spatial coexpression of the Macro1 cells and identified two main gene modules (K1, K2) and one minor module using SCoexp. The K1 module was more active in macrophages from tumor distal regions (Supplementary Fig. 10e) and correlated with several *C1Q* genes, *HAVCR2*, *CD74*, *HLA-DRA*, etc. (Supplementary Fig. 10g). Conversely, the K2 module showed an opposite spatial pattern (Supplementary Fig. 10f) and correlated with *CHIT1*, *CSTB*, *APOC1*, *MARCO* and others (Supplementary Fig. 10g).

Fig. 5 | CellTrek displays the spatial tumor-immune microenvironment in DCIS2. a, H&E image of the tissue section from the DCIS2 patient. Histopathological annotations of tumor regions are highlighted in red circles with labels from T1 to T11. **b**, UMAP of DCIS2 scRNA-seq data (tumor cells, B cells, NK/T cells, myeloid and pDC cells). **c**, CellTrek spatial mapping of tumor cells, B cells, NK/T cells, myeloid and pDC cells. Yellow boxes highlight potential locations of tertiary lymphoid structures (TLS) with aggregation of mixed immune cells. **d**, ST spot-level TLS signature scores. **e**, Boxplot showing the association between CellTrek-based immune cell counts and ST spots in TLS score quantiles (each quantile contains 61-62 ST spots). We performed a Spearman correlation test. **f**, CellTrek spatial mapping of different T cell states. The contour plot represents the tumor cell densities. **g**, UMAP of scRNA-seq data showing different T cell states. **h**, Spatial colocalization graph of T cell states using SColoc. **i**, CellTrek spatial mapping of the T exhaustion scores. **j**, UMAP of T cells showing the exhaustion scores. **k**, UMAP of T cells showing the spatial K-distances to their 15 nearest tumor cells. **l**, Boxplot comparing the T cell exhaustion scores between different T cell states (CD8T n = 455; NaïveT n = 144; CD4T n = 714; CD4Te n = 97; Treg n = 313; CD8Te n = 158). **m**, Boxplot comparing the T cell exhaustion scores between T cells proximal to tumor cells (TP) and T cells distal to tumor cells (TD) (TP n = 1,631; TD n = 250). **n**, Boxplot comparing the T cell exhaustion scores between TP and TD in each T cell state. In **l**, **m** and **n**, two-sided Wilcoxon rank sum tests were performed. Boxplots show the median with interquartile ranges (25–75%); whiskers extend to 1.5× the interquartile range from the box.

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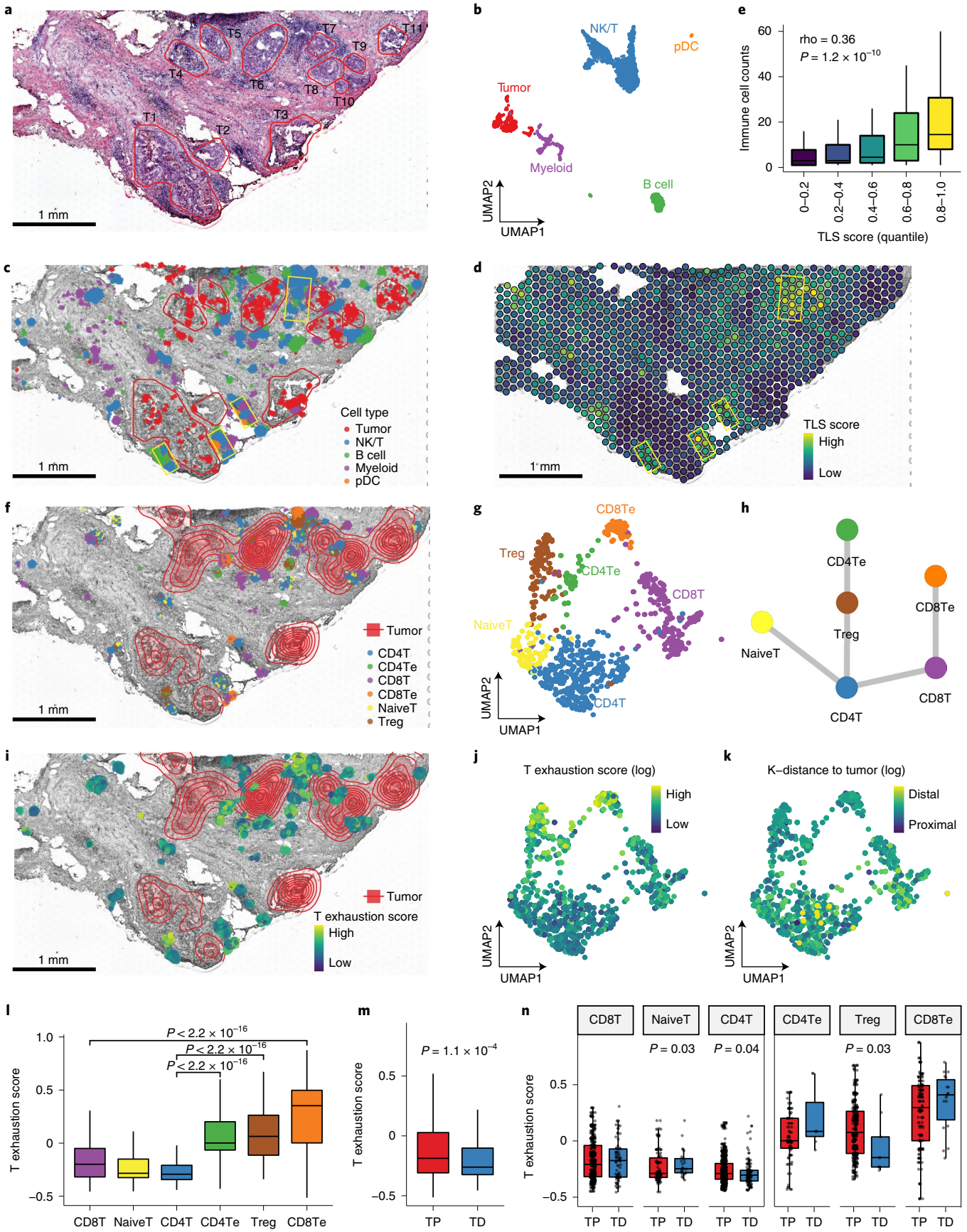

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To orthogonally validate the spatial distribution of tumor and immune cells inferred by CellTrek, we performed immunofluorescence (RNAscope) experiments with targeted probes for tissue slides from DCIS2 and another DCIS sample (DCIS3). This data showed that the DCIS tumor areas had high expression of *ERBB2*, while *TAGLN* marked the basal epithelial layers of the ducts (Supplementary Fig. 11a,b). Furthermore, immune suppressive T cell markers, including *CTLA4* and *FOXP3*, had high expression near the DCIS areas in DCIS2 (Supplementary Fig. 11b,c) which is consistent with the CellTrek results. Similarly, in DCIS3, we found immunosuppressive T cells with *CTLA4* and *FOXP3* near the ducts (Supplementary Fig. 11d–f). Additionally, this data showed that B cells (*MS4A1)*, monocytes/macrophages (*CD68*) and dendritic cells (*CD1C*) were also near the DCIS ductal regions, suggesting the presence of TLS (Supplementary Fig. 11g), and was consistent with the CellTrek results for DCIS2. In contrast, fewer immune cells were observed in the normal lobular epithelial areas in the same tissue section, particularly for the immune suppressive T cells (Supplementary Fig. 11h,i). These data confirmed our findings on the DCIS tumor-immune microenvironment that were inferred using CellTrek.

Discussion

Here, we report a computational tool, CellTrek, for reconstructing a spatial cellular map based on scRNA-seq and ST data. In contrast to conventional deconvolution approaches[6,11–13], CellTrek directly projects single cells to their spatial coordinates in tissue sections and therefore takes full advantage of the scRNA-seq data. We also developed two downstream computational modules (SColoc and SCoexp) to further analyze CellTrek results. By reconstructing a cellular spatial map, CellTrek provides several advantages. First, it provides a flexible way to investigate any feature of individual cells (for example, cell type/state, pseudotime) in a spatial manner, while most ST deconvolution approaches can decompose spots only into cell types, and cannot achieve single-cell-level feature mapping. Second, CellTrek is very flexible and can take any cell-location probability/similarity matrix as an input to reconstruct a cellular map, thus enabling further downstream analyses. Third, by using a metric learning approach and a nonlinear interpolation, CellTrek allows more accurate cell charting at higher spatial resolution. Finally, with the development of higher spatial resolution sequencing technologies, CellTrek is fully capable of charting single cells to other spatial sequencing data to provide even higher spatial granularity.

We first benchmarked the CellTrek performance using the simulated and in situ datasets and then evaluated the accuracy and robustness under different data conditions. By applying the CellTrek toolkit to two 'well-established' datasets from mouse brain and kidney, we demonstrated its capability in recovering the topological structures of different cell types. We further showed that CellTrek can identify high-resolution substructures by mapping categorical (cell states) and continuous (pseudotime) features to the tissue sections. SColoc can reconstruct the spatial relationships of different cell types into a graph, which can be further used for cell–cell communication analysis. Moreover, SCoexp can detect spatial coexpression modules in several cell types, showing topological patterns in the tissue sections.

In our study, we performed matched scRNA-seq and ST experiments of two DCIS samples. We then applied the CellTrek toolkit to delineate the spatial distribution of the tumor subclones in different ductal regions and the topological organization of the tumor-immune microenvironment. In DCIS1, we found that three tumor subclones were localized to different ducts with different levels of clonal diversity. Although morphological and genomic intratumor heterogeneity have been observed previously, here we report spatial heterogeneity in the ductal network in the DCIS tissue[42,45–47]. In DCIS2, CellTrek accurately mapped tumor and immune cells, and indicated the presence of TLS enriched with immune cells near the DCIS regions. Further analyses of T cells and myeloid cells revealed their spatial localization relative to the tumor cells. These findings were orthogonally validated using RNAscope.

While CellTrek is a powerful tool for analyzing scRNA-seq and ST data, it has several notable limitations. First, CellTrek can have sparse cell mapping in some tissue areas, as we showed in the simulation data. To overcome this problem, one can collect tissues with higher cellular density for ST analysis, sequence more cells or integrate several scRNA-seq datasets. Second, CellTrek maps cells to their most similar spots based on a sparse graph, which requires ST spots with relative high cell purities. A simulation of increasing spatial randomness (decreasing ST spots purities) showed that CellTrek could potentially oversimplify the spatial complexity for 'less-organized' tissue structures. Finally, there is a risk of overinterpreting the data based only on CellTrek since it is a computational inference tool. Although relatively stringent parameters are used as a default to control for false positives, orthogonal validation is recommended to confirm biological findings.

In the future, CellTrek could be improved by including image recognition or deep learning approaches for cell segmentation and identification. Additionally, epigenetic regulation is of great interest in developmental biology and cancer research. Therefore, another future direction is to adapt CellTrek for epigenome data (for example, scATAC-seq) to understand spatial epigenetic regulation in the tissue sections. Overall, we expect that CellTrek will have a multitude of applications for studying basic biology and human disease in a spatial context, as applying scRNA-seq and ST experiments to the same tissues is becoming ever more commonplace.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41587-022-01233-1.

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Methods

CellTrek toolkit. *CellTrek.* Using ST and scRNA-seq data, CellTrek first uses a reference-based coembedding approach from the Seurat package[24] with scRNA-seq data as the reference and the ST data as the query. From the coembedded data, CellTrek then trains a multivariate RF model with a default of 1,000 trees on ST data using *rfsrc* from the randomForestSRC package[48] with the following formula:

$$[\mathrm{X},\ \mathrm{Y}] \sim \mathrm{PCs},$$

where X and Y are spatial coordinates of ST spots, and PCs are the top principal components (default = 30). Additionally, CellTrek introduces a nonlinear two-dimensional (2D) interpolation approach from the akima package to augment the ST spots. The trained RF model is applied to the ST-scRNA coembedding data to produce an RF-based distance matrix. RF distance is calculated by

$$\text{RF distance} = \frac{\text{number of edges to the immediate shared ancestor on the tree}}{\text{number of edges to the tree root}}$$

This distance metric provides a semiparametric measurement of the similarities between data points in their feature space while supervised by the spatial coordinates. Based on the RF distance matrix, CellTrek further constructs a sparse graph by filtering distances larger than a threshold and matching the closest cell-spot pairs using MNN. This sparse graph enables a flexible cell charting scheme with certain degrees of redundancy considering that one ST spot contains several cells, and different ST spots could consist of similar cells. In addition to the default machine learning-based distance matrix, CellTrek is designed to be compatible with any cell-location similarity/probability matrix computed from other methods as an input, such as novoSpaRc[20], thus extending the compatibility and scalability of CellTrek. Using the sparse spot-cell graph, CellTrek assigns the coordinates from ST spots to their connected neighboring cells. To avoid the cell clumping problem, CellTrek applies point repulsion algorithm using *circleRepelLayout* from the packcircles package. Additionally, we also developed an interactive plotting function (*celltrek_vis*) for the CellTrek cell map visualization which allows mapping any continuous or categorical cell features to the spatial map with different colors and shapes provided. This plotting function also allows interactive manual selection and annotation of cells.

SColoc. To recapitulate the colocalization of different cell types on the CellTrek results, we developed the SColoc module that provides three different approaches to calculate cell closeness, that is, KL, DT and KD, considering different tissue structures or study goals. For the KL-based approach, SColoc calculates a 2D grid kernel density for each cell type using *kde2d* from the MASS package with default h equal to spatial distance between two neighbor ST spots and $n = 25$. Then, SColoc calculates the KL-divergence on the 2D density between each pair of two cell types. KL-based SColoc works well for detecting the global closeness of large spatial structures, for example, the neuron layer structure in the brain. For the DT-based approach, it first builds a 2D Delaunay triangulation network using *delaunayn* from the geometry package based on the CellTrek result. To confine the network complexity, SColoc can further filter edges with distances larger than a certain threshold. Then, on the cell type level, SColoc calculates neighboring cell counts on the DT network. Between any pair of cell types, SColoc calculates a log odds ratio that represents the colocalization of these two cell types. This approach shows better performances in capturing connections between cell types when local cellular structures are more of interest (for example, DCIS samples). To further simplify the connections, SColoc will also build a MST. The above calculation procedures are performed repetitively on bootstrap samples (default 20 iterations). Both bootstrap closeness matrix and MST consensus matrix are produced for graph visualization. SColoc also provides an interactive visualization function (*scoloc_vis*) to render a graphical representation of cell colocalization using either MST consensus or bootstrap closeness as input with flexible tuning thresholds to simplify the complexity.

SCoexp. To identify potential spatial-relevant gene coexpression modules, especially in cell types of interest, we developed the SCoexp module based on the CellTrek results. For cells of interest, SCoexp first calculates a spatial distance matrix between each cell and converts it into a spatial kernel matrix W using radial basis function (RBF) with a default sigma equals to the distance between two neighbored ST spots. Using this spatial kernel matrix W and a cell gene expression matrix X, SCoexp calculates a spatial-weighted gene cross-correlation matrix based on the following formula:

$$\mathrm{WCor} = \frac{X^T W X}{\sqrt{\mathrm{diag}(X^T W X)\mathrm{diag}\,(X^T W X)^T}},$$

where diag() is the diagonal vector of a matrix. Based on the spatial-weighted gene cross-correlation matrix, SCoexp uses two coexpression module detection approaches, that is, CC[21] and WGCNA[22]. The CC approach applies the ConsensusClusterPlus package with a default K through 2 to 8 and default repetition of 20. Then, for the identified gene clusters, a within-cluster filtering step removes low consensus and low correlation genes in each module. WGCNA approach identifies coexpression modules with the normalized *WCor* matrix as an input. Similarly, a within-module filtering can be applied to remove low correlation genes. For the identified gene modules, we applied the Seurat *AddModuleScore* to calculate a cell-level module activity score that can be investigated on the CellTrek spatial map.

Mouse data acquisition and analysis. *Mouse brain data.* For the mouse brain scRNA-seq (Smart-seq2)[17] and ST data (10x Genomics Visium), we downloaded the Seurat objects from https://satijalab.org/seurat/articles/spatial_vignette.html. For the scRNA-seq data, we randomly subsampled 8,000 cells from the original dataset and performed a standard data analysis procedure including log normalization, scaling, variable genes selection ($n = 5{,}000$) using *vst*, dimension reduction using principal component analysis (PCA) and UMAP. On each cell type, we then applied *dbscan* (v.1.1–5)[49] with minPts = 20 and eps = 0.5 to filter out outliers in the UMAP. For the ST data, we performed similar analysis processing (log normalization, scaling, variable genes identification and dimensionality reduction) as the scRNA-seq data.

Mouse hippocampus data. The mouse hippocampus scRNA-seq data[30] was downloaded from the Seurat website (https://satijalab.org/seurat/articles/spatial_vignette.html#slide-seq-1). We subsampled 20,000 cells from the data and performed *SCTransform*. Mouse hippocampus Slide-seq data[29] was installed through the Seurat package. We subsampled 10,000 spots and performed *SCTransform*. For the Slide-seq data, we then conducted dimensionality reduction using PCA, UMAP and clustering analysis using Seurat.

Mouse kidney data. For the mouse kidney 3′ scRNA-seq data (10x Genomics), we downloaded the filtered gene count matrices from GEO (GSE129798), and the cell type annotation from the original paper[18]. The data included single cells from three different dissection zones, that is, cortex, outer medulla and inner medulla. We processed the scRNA-seq data following procedures similar to those used for the mouse brain data. We also subsampled 8,000 cells and filtered out outlier cells using group-wise *dbscan*. On proximal tubule and distal tubule cells, we conducted trajectory analysis using the Monocle2 package (v.2.14.0)[26] with raw count data as an input and the negative binomial distribution as the expressionFamily argument. Highly variable genes with mean expression more than 0.25 were used to perform gene ordering and dimensionality reduction based on the *DDRTree* function. A principal graph was generated and cells were ordered along the pseudotime trajectory. The mouse kidney ST data was downloaded from the 10x Genomics website (https://www.10xgenomics.com/resources/datasets/) and analyzed by the same procedure as described for the mouse brain ST process workflow.

DCIS tissue collection and sequencing. *ScRNA-seq experiments.* The two DCIS samples were obtained from the MD Anderson Cancer Center. The study was approved by the Institutional Review Board (IRB) and tissue was procured with informed consent from patients. The tumors were stained with H&E and evaluated by pathology, in which DCIS1 was classified as a pure DCIS sample and DCIS2 as a synchronous DCIS-IDC tissue. The estrogen receptor (ER) and progesterone receptor (PR) status of the samples were determined by immunohistochemistry (IHC), which showed that DCIS1 was ER and PR positive, while DCIS2 was ER positive and PR negative. We used our previous protocol to prepare viable single-cell suspensions[40]. Briefly, fresh tissue samples were dissociated into viable single-cell suspensions by enzymatic dissociation using collagenase A and trypsin. The fresh tissues were also embedded into optimal cutting temperature compound (OCT) and snap-frozen. The viable cell suspensions were used as input material for scRNA-seq using the Single-Cell Chromium 3′ protocol by V2 (10x Genomics, catalog no. CG00052, PN-120237) and V3 (10x Genomics, catalog no. CG000183, PN-1000075) chemistry reagents. The final libraries containing barcoded single-cell transcriptomes were sequenced at 100 cycles using the S2 flowcell on the Novoseq 6000 system (Illumina).

ST Visium experiments. Fresh tissues from the two DCIS patients were cut to the appropriate size and embedded in cryomold (Fisher, catalog no. NC9542860) by OCT (Fisher, catalog no. 1437365) on dry ice and stored in −80°C in sealed bags. Frozen OCT-embedded DCIS cryosections were cut to 12 µm in the cryostat (Thermo Scientific Cryostar, catalog no. NX70) with specimen head temperature at −17 °C and blade temperature at −15 °C. The cut sections were placed in a capture area of the Visium spatial slide (10x Genomics, catalog no. PN-1000184). The slide was permeabilized for 12 min according to the Visium Spatial Tissue Optimization protocol (10x Genomics, catalog no. CG000238). We performed imaging of the stained slides using the Nikon Eclipse Ti2 system. Finally, the ST libraries were constructed by following the Visium Spatial Gene Expression protocol (10x Genomics, catalog no. CG000239) and sequenced at 200 cycles by S1 flowcell on the Novoseq 6000 system (Illumina).

RNAscope experiments. RNAscope probes were ordered from Advanced Cell Diagnostics for the following genes: *ERBB2*, *ACTG2*, *TAGLN*, *CTLA4*, *FOXP3*, *CD3D*, *CD4*, *CD8A*, *MA4A1*, *CD1C* and *CD68*. Cryosections of two snap-frozen

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DCIS samples (OCT embedded) were cut at a thickness of 10 µm and used to performed RNA in situ hybridization assay with the RNAscope Multiplex Fluorescent v.2 kit according to the manufacturer's instructions (catalog nos. 323110 and 323120) with following modifications: tissue sections were fixed in 10% neutral buffered formalin for 1 h at 4°C, all washing steps were increased to three times (3–5 min each wash), areas enclosed by hydrophobic barrier were 1.905 cm × 1.905 cm, for all reagent steps 150 µl were dispensed, tissues were treated with Protease III for 15 min at room temperature, Opal dyes were used at 1/750–1/2250 (Akoya Biosciences, catalog nos. FP1487001KT, FP1488001KT, FP1495001KT FP1497001KTP), kit 4′,6-diamidino-2-phenylindolestain was replaced by a 1/2,000 working stock in PBS (Invitrogen, catalog no. D1306 in DMF, 5 mg ml^{-1}), and slides were mounted in Prolong diamond (Invitrogen catalog no. P36970). Fluorescent Images were scanned using a motorized stage on the Nikon Eclipse Ti2 microscope with a ×20 objective and analyzed with the Nikon NIS-Elements AR software (v.5.30.04).

DCIS scRNA-seq data analysis. FASTQ files were first preprocessed using the *Cell Ranger* v.3.1.0 pipeline (10x Genomics) with default arguments and mapped to the *GRCh38* reference genome to construct count matrices. Unique molecular identifier (UMI) counts were then processed using the Seurat package (v.3.2)[24]. For the two DCIS samples, cells with fewer than 700 unique feature counts were filtered. We also filtered cells that had mitochondrial counts of greater than 15%. Counts were then normalized using the *NormalizeData* with default *LogNormalize* method. Afterwards, normalized counts were scaled and centered using *ScaleData* function. A total of 2,000 variable genes were found using *FindVariableFeatures* function and PCA was conducted using *RunPCA* with default parameters. *ElbowPlot* was used to determine the number of principal components (PCs) for the downstream analyses. Ten neighbor cells and the top 20 PCs were used for neighbor finding using *FindNeighbors*. Cell clusters were identified using *Louvain* algorithm of *FindCluster* function with the top 20 PCs and a resolution of 0.6. We ran *RunUMAP* to visualize cell manifolds in a 2D space. DE genes were identified using *FindAllMarkers*. Cell identities were annotated based on a combination of two strategies: (1) top markers for each cluster based on the DE gene analysis and (2) Canonical cell type specific marker expression using *FeaturePlot*. We used dbscan with minPts = 5 and eps = 0.5 to remove outliers for each cell type based on the UMAP. Then, we utilized CopyKAT (v.17) to infer the copy number for each cell[40]. For DCIS1, the inferred copy number profiles were plotted using ComplexHeatmap (v.2.2.0)[50] and used to generate copy number alterations (CNA)-based UMAP using the uwot package (v.0.1.8) on the Manhattan distance matrix. We then applied dbscan to identify tumor subclones in a CNA-based UMAP space. Consensus copy number profiles for the tumor subclones and normal cells were then used to construct a phylogenetic tree based on neighbor-joining tree in the ape package (v.5.4)[51] and the normal cell was chosen as root. We performed gene set enrichment analysis on three tumor subclones using the fgsea package[41]. Cell cycle scores for cells were calculated using the *CellCycleScore* module from Seurat and assigned as in G1, G2/M or S phase. For DCIS2, to further define NK/T and myeloid cell subtypes, these cells were extracted and reanalyzed using the similar workflow. We performed DE analysis and cell subtypes annotated based on the top gene expression and canonical maker expression. To characterize T cell exhaustion, we used an exhaustion gene signature including *PDCD1*, *CTLA4*, *LAG3*, *HAVCR2*, *CD244*, *CD160* and *TIGIT*.

DCIS ST data analysis. Sequencing data were first preprocessed with *Space Ranger* v.1.0.0 and mapped to the *GRCh38* reference genome. Similar to the scRNA-seq data analysis workflow, the ST data were subsequently processed using Seurat. We filtered out spots with UMI counts less than 100. The UMI counts were normalized using *NormalizeData* function with *LogNormalize followed by ScaleData*. A total of 2,000 variable genes were found using *FindVariableFeatures* function and we performed PCA using *RunPCA*. For DCIS1, to study tumor spots, we selected the ST spots covering the ductal areas based on histopathology and conducted an unsupervised clustering using the top 20 PCs at resolution of 0.5. For DCIS2, we cropped out the damaged areas of the original tissue slide, which resulted in a total of 965 spots left. The downstream analyses were based on the cropped tissue section. For ST spots, the TLS score was calculated based on a 12-gene signature[44] using *AddModuleScore* function in Seurat.

Benchmarking and simulations. *Simulation data.* We used the R package Splatter[52] to generate the simulation scRNA-seq data with 6,000 cells, 5,000 genes and five cell groups. We set lib.loc = 8.5, lin.scale = 0.4 and dropout.type = 'group'. We used the function *splatSimulatePaths* to simulate a sequential manifold with path.skew = 0.1, de.facLoc = 0.1 and de.facScale = 0.8 for all groups. We then generated a customized spatial area and assigned cells to the locations using optimal transport between the top two PCs and spatial locations using the transport package[53]. ST spots were generated using a 30 × 20 grid and spots outside the spatial area were then dropped. Each ST spot aggregated the gene expression from its five nearest cells. A final ST data with 394 spots was used.

Drosophila embryo data. This data is from the Berkeley *Drosophila* Transcription Network Project (BDTNP) (https://www.fruitfly.org/)[20]. Based on the FISH data of *Drosophila* embryo, they generated 3,309 cells with the expression level of 84 landmark genes. We filtered five outlier cells and applied the Seurat analysis pipeline to identify seven cell groups at resolution = 0.25. Similarly, we generated the ST data with a final of 483 spots.

Mouse embryo data. We downloaded the seqFISH data of mouse embryo 1 from https://marionilab.cruk.cam.ac.uk/SpatialMouseAtlas/ (ref. [23]). We downsampled 6,000 cells and removed the low-quality group and groups with fewer than 50 cells. We finally used a single-cell data of 5,852 cells and 351 genes for test. ST data was correspondingly generated with a final of 779 spots. We subselected a group of foregut cells and ran Monocle2 for trajectory analysis.

Different conditions of simulation. Based on the original scRNA-seq simulation data, we further simulated different data conditions to test the robustness of CellTrek. In simulation-1, we simulated different read counts for both scRNA-seq and ST data by dropping a certain number of reads from a proportion of cells and spots. The detailed dropout parameters were described in Supplementary Table 1. This simulation will not only produce data with different library size and read depth but also introduce batch effects between scRNA-seq and ST data since the dropouts were conducted independently on these two datasets. In simulation-2, we simulated spatial randomness by swapping cells in a spatial manner. Specifically, we first added a Gaussian noise with a customized deviation to the original cell coordinates and then assigned the cells with spatial noise back to the original tissue structure (Supplementary Table 2 and Supplementary Fig. 4c). The increasing Gaussian noise deviation will increase the spatial randomness compared with the raw reference structure (Supplementary Fig. 4d, left). Corresponding ST data were then generated at each condition. This simulation also produced a decrease of cell type purities within ST spots (Supplementary Fig. 4d, right). In simulation-3, we simulated different tissue densities by spatially downsampling cells. We first set up some spatial marker points (Supplementary Table 3 and Supplementary Fig. 4g), and then calculated cell distances to the marker points and converted into RBF kernel densities. We then downsampled cells based on their kernel densities. This simulation can generate different spatial sparseness in different areas. ST data was then generated correspondingly. To quantitatively evaluate the CellTrek performance, we compared the CellTrek results with their spatial cellular reference under different conditions using cell type level KL-divergences and correlation analysis on spatial coordinates. Permutation tests of 100 times were performed to generate null distributions.

CellTrek and downstream analyses. *Simulation and benchmarking data.* For the simulated scRNA-seq and ST data, we first ran *traint* to coembed the data into a shared feature space with default parameters. We tested the RF distance distribution between ST and scRNA-seq data and determined a threshold around 1% quantile which is 0.35 (350 out of 1,000). Using this threshold, we ran *celltrek* on the coembedded *traint* data with following parameters: intp_pnt = 2,000 spots, nPCs = 20, ntree = 1,000, top_spot = 4, spot_n = 5 and repel_r = 10 with ten iterations. For NVSP-CellTrek, we first ran *novoSpaRc* using reference-guided mode using the same default parameters following the protocol from https://github.com/rajewsky-lab/novosparc/blob/master/reconstruct_drosophila_embryo_tutorial.ipynb. On the *gw* matrix from *novoSpaRc*, we ran a negative log transformation and determined a threshold of 13 (around 1% quantile of the distance). We then ran *celltrek_from_dist* with following parameters: top_spot = 4, spot_n = 5, dist_cut = 13, and reprel_r = 20. For SrtCT, we ran *FindTransferAnchors* from Seurat with ST data as the reference and scRNA-seq data as the query and reduction of 'cca'. We then applied *TransferData* to transfer the ST coordinates from ST data to single cells with default parameters. For *Drosophila* and mouse embryo data, similar to the simulation data, we performed CellTrek, NVSP-CellTrek and SrtCT to analyze the single cell and generated ST data using similar parameter settings. For CellTrek simulation analysis of different data conditions, we first filtered low-quality ST and scRNA-seq data using nFeature >10. We set intp = FALSE due to the running time and repel_r was set to 20 considering no spatial interpolation here. All parameters were fixed across different simulations and conditions.

Mouse brain. We subset the frontal cortex region in the ST data for single-cell spatial charting. For the normalized ST and scRNA-seq data, we first ran *traint* to coembed the data into a shared feature space with default parameters. We tested the RF distance distribution between ST and scRNA-seq data and determined a threshold close to 1% quantile which is 0.5 (500 out of 1,000). Using this threshold, we ran *celltrek* on the coembedded *traint* data with following parameters: intp_pnt = 10,000 spots, nPCs = 20, ntree = 1,000, top_spot = 5, spot_n = 10 and repel_r = 20 with ten iterations. For NVSP-CellTrek, we ran *novoSpaRc* using reference-guided mode. Similarly, we ran a negative log transformation on the *gw* matrix and determined a threshold of 14 by testing the distribution (1% quantile). We then ran *celltrek_from_dist* with following parameters: top_spot = 5, spot_n = 10, dist_cut = 14 and reprel_r = 20. We performed SrtCT with the same parameters as the simulation study. To evaluate different cell charting approaches, we benchmarked SrtLT as our reference. Specifically, we used *SelectIntegrationFeatures* to select 2,000 features between ST and scRNA-seq data.

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We then used *FindTransferAnchors* with scRNA-seq data as the reference, ST data as the query data and reduction of 'cca'. Next, we applied *TransferData* to transfer the cell type labels from scRNA-seq data to the ST spots. For comparison, for each cell type, we converted all charting results (CellTrek, NVSP-CellTrek and SrtCT) to a spatial grid density with h = 10, $n = 50$ and coordinates limits fixed to the range of ST spots coordinates. For SrtLT, we first applied a customized function that generates pseudopoints based on probabilities and used the same approach to calculate a spatial grid density for each cell type. We then focused on cell types that had (1) more than ten spots with label transfer probability >0.5 and (2) more than 20 cells in the scRNA-seq data. For each cell type, we used total normalization and calculated the KL-divergence between any cell chart density and the SrtLT reference density. To calculate the cell spatial colocalization, we applied *scoloc* on the CellTrek results using the KL approach with cell_min = 15, eps = 1×10^{-50} and boot_n = 20. We calculated the mean of bootstrap KL matrices and converted it into a similarity matrix. This matrix was then used for heatmap with *ward.D2* clustering. For the graph visualization, we used the MST consensus matrix as input while setting a consensus cutoff at 0.3. For the glutamatergic neurons, we calculated spatial K-distance by setting the L2/3 IT cell type as the reference and k = 10. K-distance was then normalized by the maximum values. For L5 IT cells, we performed the spatial coexpression analysis. We first filtered mitochondria, ribosomal and highly sparse (nonzero proportion <20%) genes and calculated highly variable genes, which resulted in a total of 2,000 genes. We then used a consensus clustering based spatial coexpression analysis using *scoexp* with following parameters: sigm = 140, avg_con_min = 0.5, avg_cor_min = 0.4, zero_cutoff = 3, min_gen = 50, max_gen = 400 and maxK = 8. After gene modules were identified, we applied *AddModuleScore* from Seurat to calculate the cell module activity score with nbin = 10 and ctrl = 20. We performed gene ontology (GO) enrichment analysis for the two modules using the clusterProfiler package[54]. To identify genes correlated with the coexpression modules, we used a customized *FindCorMarkers* function in CellTrek based on the Spearman correlation.

Mouse hippocampus. For the Slide-seq and scRNA-seq on mouse hippocampus, *traint* and *celltrek* were consequentially conducted with norm = 'SCT', *n*PCs = 50, dist_thresh = 0.65, top_spot = 5, spot_n = 2, and repel_r = 10.

Mouse kidney. Similar to the mouse brain data, we performed three different cell charting approaches for comparison, that is, CellTrek, NVSP-CellTrek and SrtCT to analyze this scRNA-seq and ST data. For CellTrek, the dist_thresh was set to 0.6 considering the distance distribution. For NVSP-CellTrek, we selected a distance cutoff of 13.5. We also benchmarked SrtLT and compared these cell charting results using KL-divergence on the spatial grid density at h = 20 and $n = 40$. For the spatial colocalization analysis, we used *scoloc* with the DT approach on the CellTrek results. The diagonal elements of the mean matrix were excluded to emphasize only the colocalization between different cell types. The MST consensus matrix was employed for graph visualization at a cutoff of 0.2. Similarly, the similarity matrix was used for heatmap with the *ward.D2* clustering. We applied *scoexp* with avg_cor_min = 0.3 on the distal tubule cells to identify the spatial coexpression modules. We also performed GO enrichment analysis for these gene modules. To evaluate cellular interactions between different cell types of mouse kidney, we applied CellChat (v.1.1.3)[55] to infer ligand-receptor interactions from the scRNA-seq data. We used the normalized count data as an input and followed the CellChat tutorial with default parameters and CellChatDB.mouse as the interaction database. Cellular interactions were visualized using *netVisual_circle* function. Next, to obtain a more specific interaction results, we used our spatial SColoc graph. We binarized the MST consensus matrix at 0.2 as a spatial weight matrix. We calculated the element-wise product of CellChat cellular interaction matrix and the weight matrix. Ligand-receptors interaction examples were then plotted using *netVisual_chord_gene* function.

DCIS1. We applied *traint* followed by *celltrek* with dist_thresh = 0.5, top_spot = 5, spot_n = 10, and repel_r = 5 to construct a spatial cell map. To calculate the diversities of different ducts, we assigned the tumor cells to their closest ducts with spatial distances less than 60. The Shannon index was then calculated using *entropy.empirical* from the entropy package and rescaled to 0–1. To identify the similarity between the ST spots of the annotated ducts and the tumor subclones identified from the scRNA-seq data, we conducted DE analyses for the ST clusters and tumor subclones, respectively. Genes with adjusted *P* values (Bonferroni correction) less than 0.1 were selected to perform Pearson's correlation between these two modalities on average log fold-change values. Spatial coexpression was conducted using the *scoexp* 'cc' approach with sigm = 60, avg_cor_min = 0.4, zero_cutoff = 3, min_gen = 50 and max_gen = 400. Similarly, GO enrichment analysis was also performed for three spatial gene modules.

DCIS2. We used the same CellTrek procedure as described for DCIS1. To test the association between ST-TLS scores and CellTrek immune cell numbers (that is, NK/T, B, myeloid and pDC), we first assigned the immune cells to their closest ST spots and counted the cell numbers. Next, we calculated the Spearman correlation between spot-level TLS scores and the corresponding immune cell counts. Within the T cell group, we applied *scoloc* with the DT approach to summarize cell colocalization. The MST consensus was used for graph visualization. We calculated the K-distance of T cells to the tumor cells with K = 15. The K-distances of the T cells were then binarized to TD and TP groups based on a Gaussian mixture model[55]. For myeloid cells, we applied the same analysis procedure as T cells for the cell colocalization and K-distance analysis. For macrophages, we also conducted a spatial coexpression analysis using *scoexp* and identified highly associated genes using *FindCorMarkers*.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The scRNA-seq and ST data were submitted to the Gene Expression Omnibus (GEO): GSE181254.

Code availability

The CellTrek software toolkit is available at GitHub: https://github.com/navinlabcode/CellTrek.

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Author contributions

R.W. and S.H. developed the CellTrek method, analyzed data, prepared figures and wrote the manuscript. S.B. and E.S. performed single-cell and ST experiments. M.H. processed data. A.T. and S.K. collected the DCIS tissue samples, interpreted data and managed the IRB protocols. K.C. provided input on the CellTrek method and manuscript. N.E.N. managed the project and wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41587-022-01233-1.

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No code was used to collect data
Data analysis	The custom code has been deposited to GitHub: https://github.com/navinlabcode/CellTrek. All open software has been described and cited in the manuscript. Data processing and analysis were done using CellRanger v3.1.0, SpaceRanger v1.0.0, Nikon NIS-Elements AR v5.30.04, R 3.6.2, and R packages Seurat v3.2, Monocle v2.14.0, CopyKAT v17, fgsea v1.12.0, clusterProfiler v3.14.3 and CellChat v1.1.3.

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Policy information about availability of data

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- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

All raw single cell RNA-seq and ST data and processed gene expression matrix have been submitted to Gene Expression Omnibus (GEO): GSE181254 and are made publicly available. There will be no restrictions on data availability. Figures are plotted based on processed data.
CellChatDB.mouse is from the CellChat package.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The public single cell RNA-seq and ST data analyzed in this study were selected based on the total number of mouse brain and kidney samples available in a public dataset. Our own single cell RNA-seq data that was generated was based on the availability of surgical samples from ductal carcinoma in situ patients. This study is exploratory and does not depend on specific statistical tests to establish significance for patient numbers. Therefore power calculations are not required to establish sample size.
Data exclusions	A subset of single cell RNA-seq and ST data were excluded from the study during a QC filtering step, which excluded cells with insufficient sequencing reads or gene detection and clustering outliers. The specific filtering criteria used during these steps as stated in the methods sections.
Replication	The single cell RNA-seq and ST experiments were collected human tissue samples and the entire sample was used for analysis. Therefore technical replicates are not possible with these human tissue samples (as with experimental models or culture systems). However, we sequenced thousands of single cells and ST spots per patient sample, in which each single cell RNA and ST dataset serves as data replication and provides an estimate of the technical noise.
Randomization	There are no experimental groups in this study.
Blinding	There are no experimental groups in this study.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Human research participants

Policy information about studies involving human research participants

Population characteristics	Two early stage DCIS breast cancer tumors are included in this study who were enrolled in a clinical protocol at the MD Anderson Cancer Center
Recruitment	The two patients recruited were both women with early stage breast cancer, and were over the age of 18
Ethics oversight	The study was approved by the IRB at the MD Anderson Cancer Center and all patients were enrolled under informed consent procedures

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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https://doi.org/10.1038/s41587-022-01233-1

Supplementary information

Spatial charting of single-cell transcriptomes in tissues

In the format provided by the authors and unedited

Spatial charting of single cell transcriptomes in tissues

Wei and He et al.

Supplementary Materials

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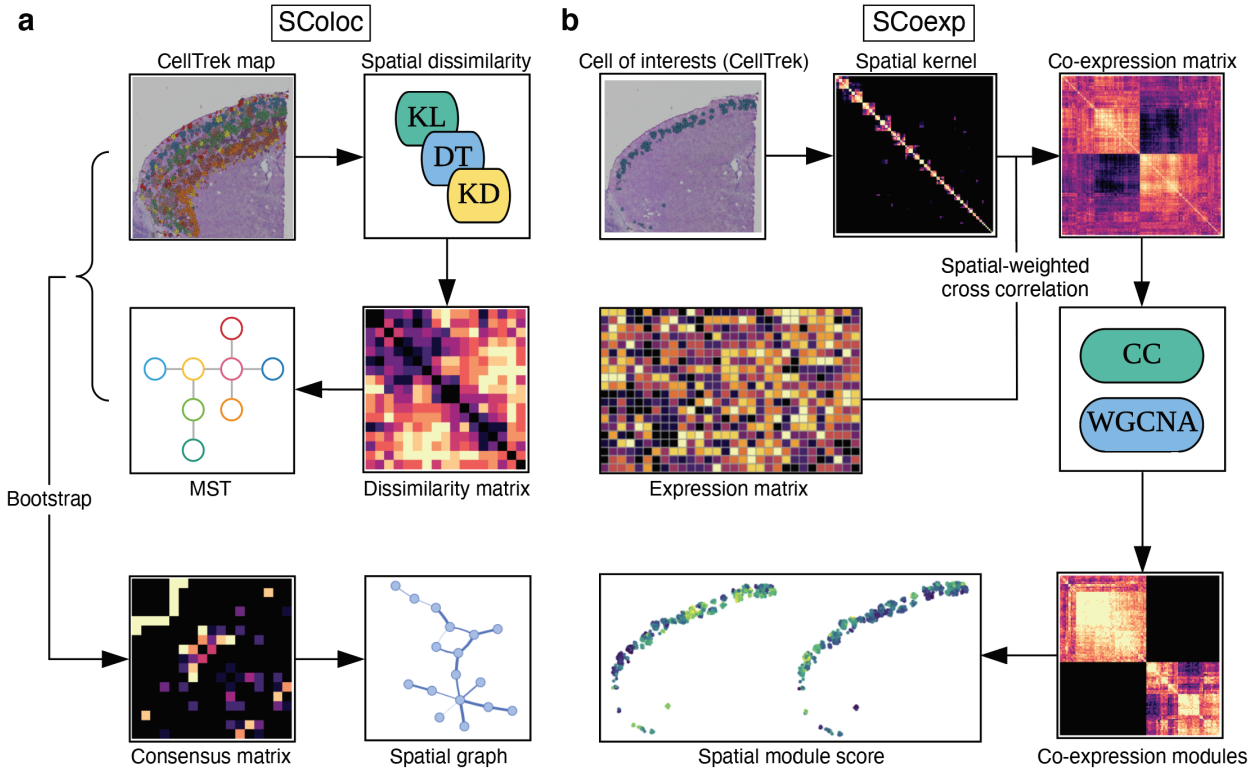
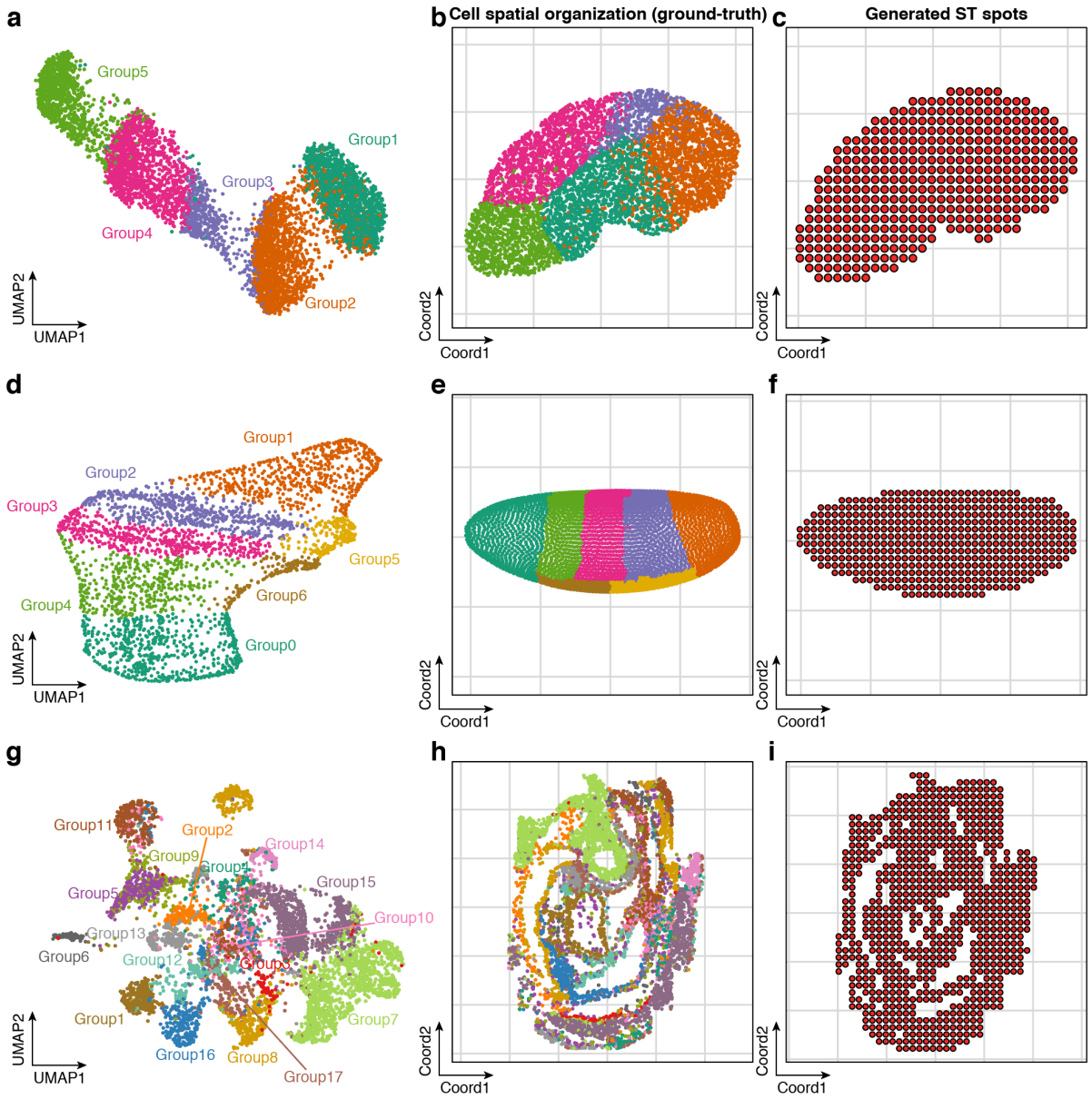


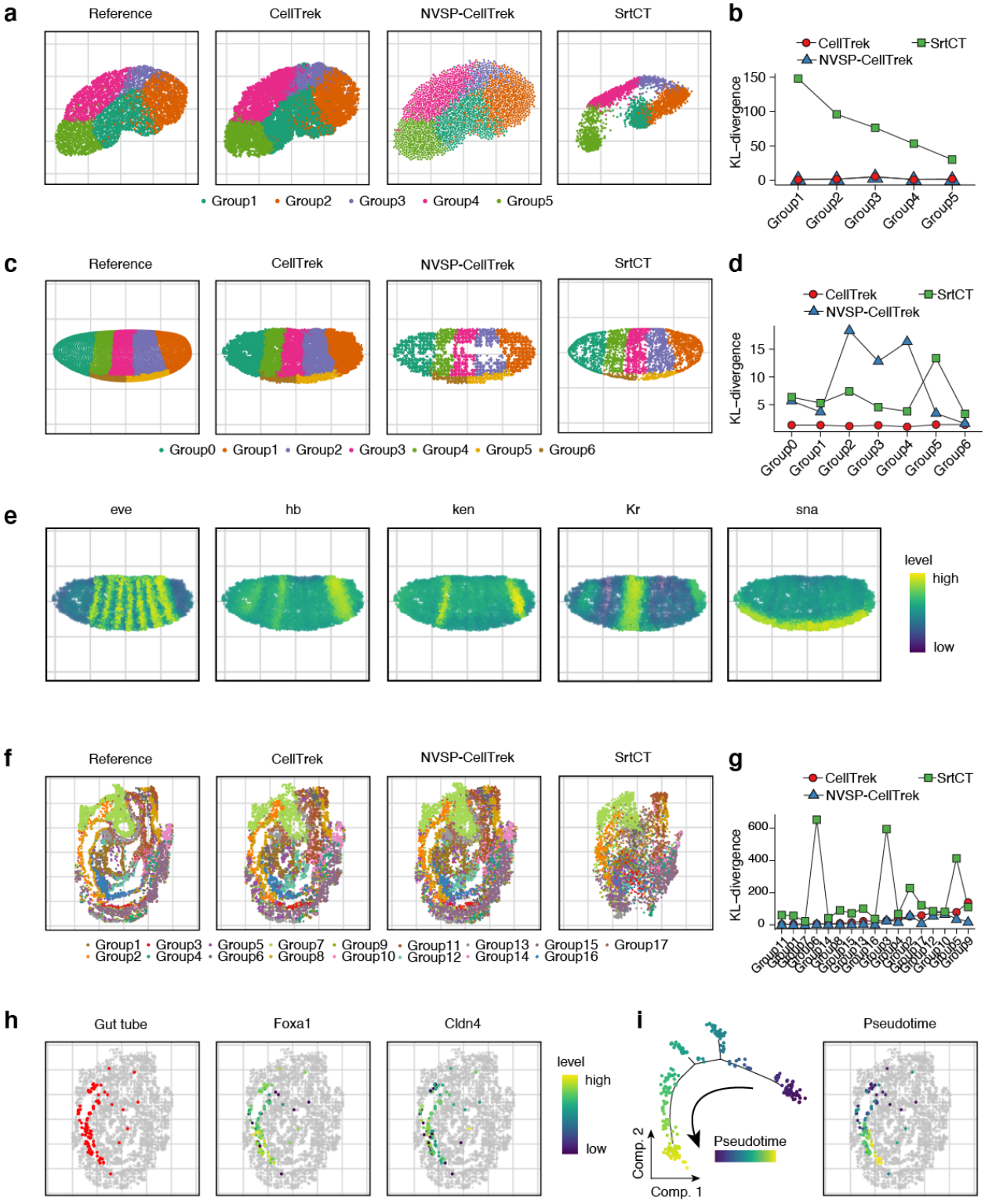
Supplementary Figure 1 – Overview of two downstream computational modules of the CellTrek toolkit

(**a**) The SColoc module. Based on the CellTrek result map, three different spatial dissimilarity methods, i.e., KL, DT, KD, can be applied to calculate a cell-type spatial dissimilarity matrix, and an MST is used to generate a tree structure. These steps are conducted repetitively on bootstrapped samples to calculate a consensus matrix on dissimilarity matrices or MSTs, which produces a final cell-type spatial graph representation. (**b**) The SCoexp module. For cells of interest, based on the CellTrek map, SCoexp first calculates a spatial kernel matrix using RBF based on their spatial distance. Next, based on the spatial kernel matrix and cell-gene expression matrix, SCoexp calculates the spatial weighted gene co-expression. Gene modules are then identified using CC or WGCNA. For the identified co-expression modules, module activity scores can be computed and mapped back to the CellTrek coordinates.



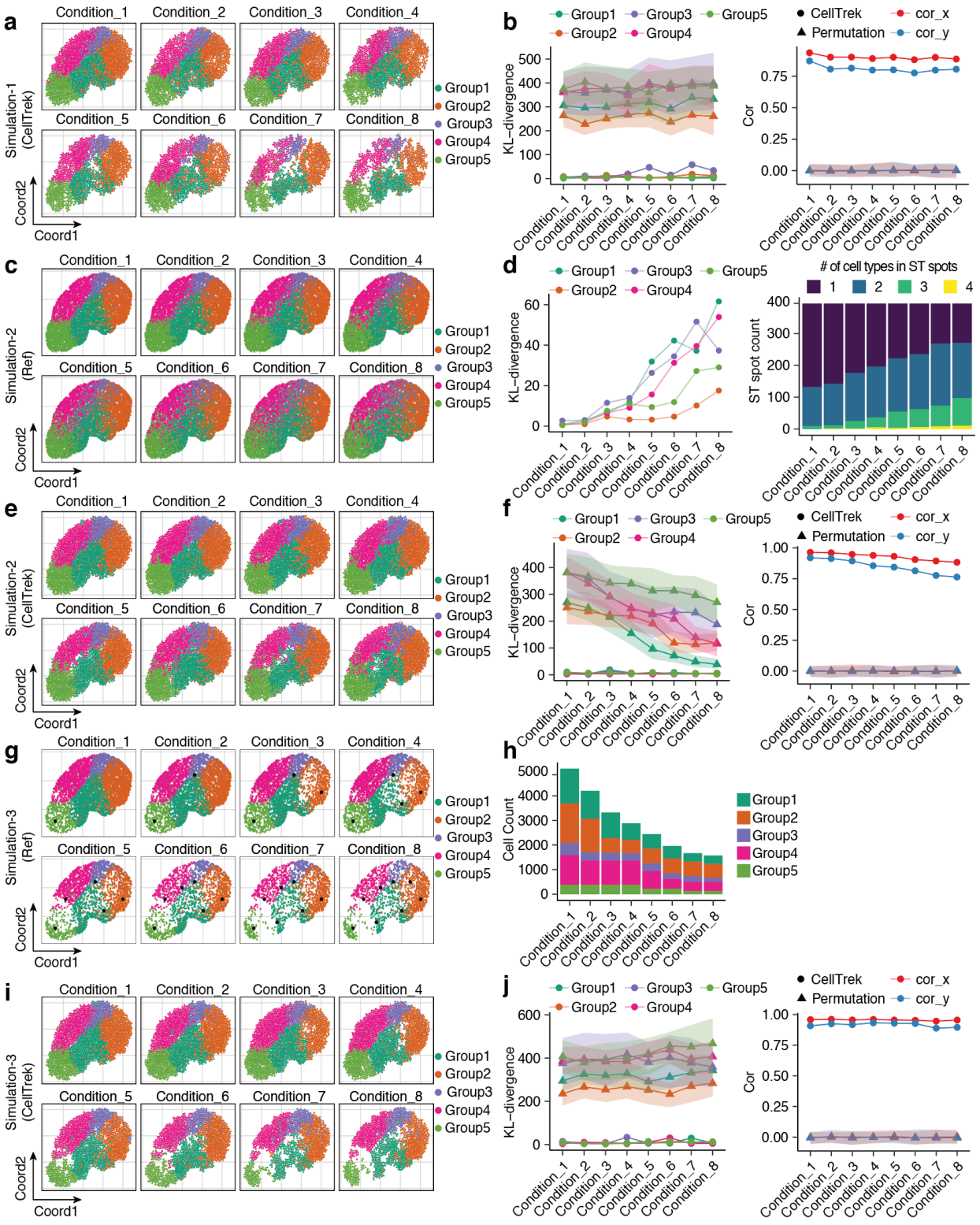
Supplementary Figure 2 – Three spatial single-cell datasets for benchmarking

(**a**) UMAP of a simulated scRNA-seq data with 5 cell groups. (**b**) Spatial organization of the simulated data as the ground truth. (**c**) ST data generated based on panel (**b**). (**d**) UMAP of *Drosophila* embryo FISH-generated single cell data. (**e**) Spatial organization of *Drosophila* embryo cells as the ground truth. (**f**) *Drosophila* embryo ST data generated based on panel (**e**). (**g**) UMAP of mouse embryo seqFISH data (Group1: Cardiomyocytes; Group2: Cranial mesoderm; Group3: Definitive endoderm; Group4: Dermomyotome; Group5: Endothelium; Group6: Erythroid; Group7: Forebrain midbrain hindbrain; Group8: Gut tube; Group9: Haematoendothelial progenitors; Group10: Intermediate mesoderm; Group11: Lateral plate mesoderm; Group12: Mixed mesenchymal mesoderm; Group13: Neural crest; Group14: Presomitic mesoderm; Group15: Spinal cord; Group16: Splanchnic mesoderm; Group17:Suface ectoderm). (**h**) Spatial organization of the mouse embryo cells as the ground truth. (**i**) mouse embryo ST data generated based on panel (**h**). In (**c**), (**f**) and (**i**), each ST spot aggregates the 5 nearest cells to generate the ST data.


a
Reference
CellTrek
NVSP-CellTrek
SrtCT
Group1 Group2 Group3 Group4 Group5
b
CellTrek
SrtCT
NVSP-CellTrek
KL−divergence
150
100
50
0
Group1 Group2 Group3 Group4 Group5
c
Reference
CellTrek
NVSP-CellTrek
SrtCT
Group0 Group1 Group2 Group3 Group4 Group5 Group6
d
CellTrek
SrtCT
NVSP-CellTrek
KL−divergence
15
10
5
Group0 Group1 Group2 Group3 Group4 Group5 Group6
e
eve
hb
ken
Kr
sna
level
high
low
f
Reference
CellTrek
NVSP-CellTrek
SrtCT
Group1 Group3 Group5 Group7 Group9 Group11 Group13 Group15 Group17
Group2 Group4 Group6 Group8 Group10 Group12 Group14 Group16
g
CellTrek
SrtCT
NVSP-CellTrek
KL−divergence
600
400
200
0
h
Gut tube
Foxa1
Cldn4
level
high
low
i
Pseudotime
Comp. 2
Comp. 1
Pseudotime

Supplementary Figure 3 – Benchmarking on three datasets with known spatial references

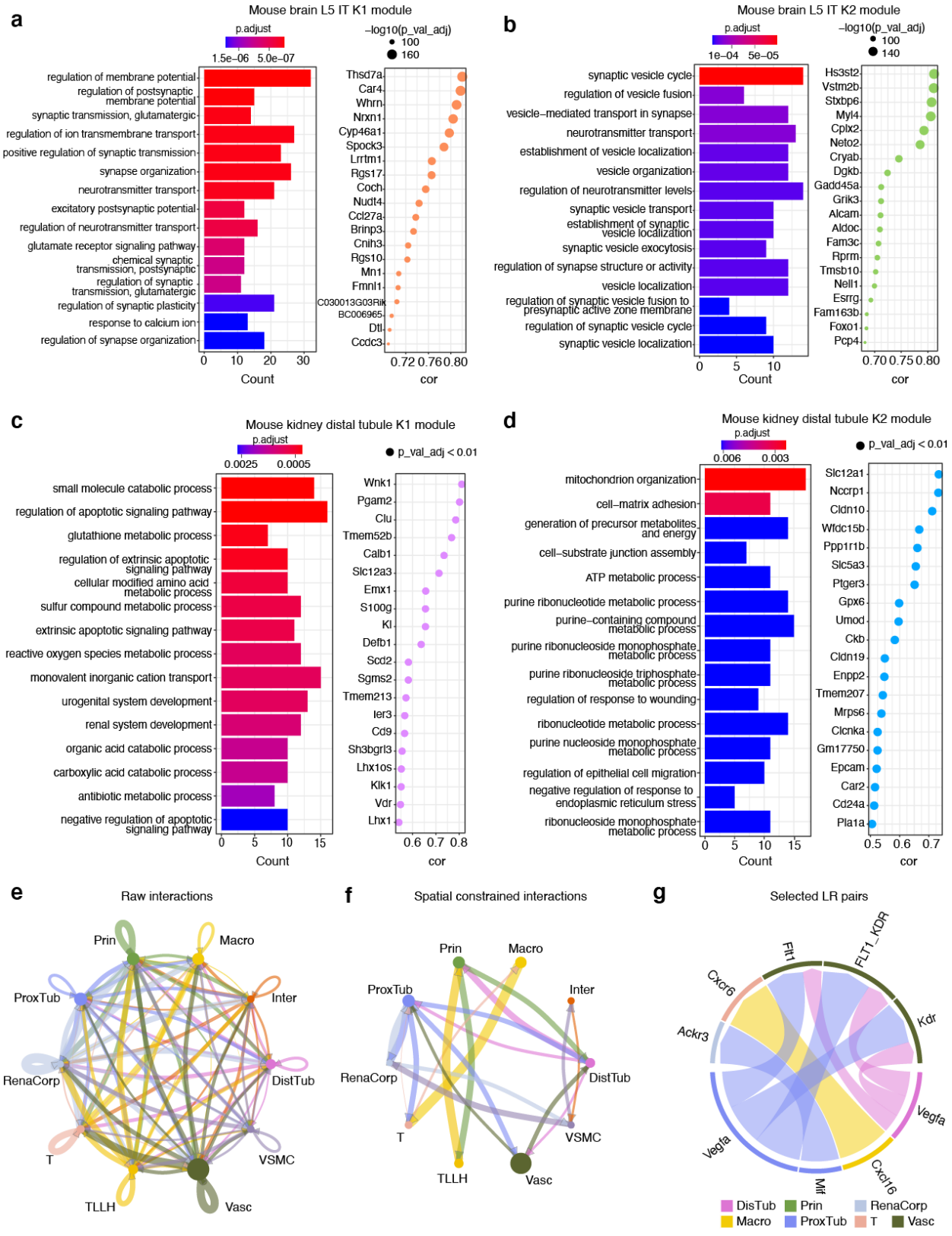
(**a**) Comparison of the ground-truth reference, CellTrek, NVSP-CellTrek and SrtCT results on the simulated data. (**b**) KL-divergence of spatial cell charting methods for each cell type compared with the ground-truth. (**c**) Comparison of the ground-truth reference, CellTrek, NVSP-CellTrek and SrtCT results on the *Drosophila* embryo data. (**d**) KL-divergence of spatial cell charting methods for each cell type compared with the ground-truth. (**e**) The expression of *Drosophola* embryogenic marker genes (*eve*, *hb*, *ken*, *Kr* and *sna*) on the CellTrek map. (**f**) Comparison of the ground-truth reference, CellTrek, NVSP-CellTrek and SrtCT results on the mouse embryo data. (**g**) KL-divergence of three spatial cell charting methods for each cell type compared with the ground-truth. (**h**) The CellTrek map of a subgroup of gut tube (foregut cells) and the marker genes (*Foxa1* and *Cldn4*). (**i**) Trajectory analysis for foregut tube cells (left) and CellTrek spatial mapping of the pseudotime values in the tissue section (right).



Supplementary Figure 4 – Simulation study of CellTrek under different data conditions

(**a**) CellTrek results under different gene counts in the simulated data. From condition1 to condition8, the gene counts decreased. (**b**) KL-divergence (left) and spatial coordinate

correlation (right) of CellTrek and permutation test across different gene count conditions. (**c**) Simulation of the increasing spatial randomness of the reference data from condition1 to condition8. (**d**) KL-divergence of the simulated reference compared with the raw spatial cell organization (left) and the number of cell types in ST spot (right) across different spatial randomness conditions. (**e**) CellTrek results under different spatial randomness of the reference data. (**f**) KL-divergence (left) and spatial coordinate correlation (right) of CellTrek and permutation test across different spatial randomness conditions. (**g**) Simulation of the decreasing tissue spatial densities (cell numbers) of the reference data from condition1 to condition8. Black dots display the marker points for spatial down-sampling. (**h**) Bar plot showing the number of cells under different tissue spatial density conditions. (**i**) CellTrek results under different tissue densities of the reference data. (**j**) KL-divergence (left) and spatial coordinate correlation (right) of CellTrek and permutation test across different tissue spatial density conditions. Detailed parameter settings are provided in Supplementary Table 1-3. For the permutation tests, the ribbon shows the 95% confidence intervals.


a
Mouse brain L5 IT K1 module
p.adjust
1.5e−06 5.0e−07
−log10(p_val_adj)
100
160
regulation of membrane potential
regulation of postsynaptic membrane potential
synaptic transmission, glutamatergic
regulation of ion transmembrane transport
positive regulation of synaptic transmission
synapse organization
neurotransmitter transport
excitatory postsynaptic potential
regulation of neurotransmitter transport
glutamate receptor signaling pathway
chemical synaptic transmission, postsynaptic
regulation of synaptic transmission, glutamatergic
regulation of synaptic plasticity
response to calcium ion
regulation of synapse organization
0 10 20 30
Count
Thsd7a
Car4
Whrn
Nrxn1
Cyp46a1
Spock3
Lrrtm1
Rgs17
Coch
Nudt4
Ccl27a
Brinp3
Cnih3
Rgs10
Mn1
Fmnl1
C030013G03Rik
BC006965
Dtl
Ccdc3
0.72 0.76 0.80
cor
b
Mouse brain L5 IT K2 module
p.adjust
1e−04 5e−05
−log10(p_val_adj)
100
140
synaptic vesicle cycle
regulation of vesicle fusion
vesicle−mediated transport in synapse
neurotransmitter transport
establishment of vesicle localization
vesicle organization
regulation of neurotransmitter levels
synaptic vesicle transport
establishment of synaptic vesicle localization
synaptic vesicle exocytosis
regulation of synapse structure or activity
vesicle localization
regulation of synaptic vesicle fusion to presynaptic active zone membrane
regulation of synaptic vesicle cycle
synaptic vesicle localization
0 5 10
Count
Hs3st2
Vstm2b
Stxbp6
Myl4
Cplx2
Neto2
Cryab
Dgkb
Gadd45a
Grik3
Alcam
Aldoc
Fam3c
Rprm
Tmsb10
Nell1
Esrrg
Fam163b
Foxo1
Pcp4
0.70 0.75 0.80
cor
c
Mouse kidney distal tubule K1 module
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p_val_adj < 0.01
small molecule catabolic process
regulation of apoptotic signaling pathway
glutathione metabolic process
regulation of extrinsic apoptotic signaling pathway
cellular modified amino acid metabolic process
sulfur compound metabolic process
extrinsic apoptotic signaling pathway
reactive oxygen species metabolic process
monovalent inorganic cation transport
urogenital system development
renal system development
organic acid catabolic process
carboxylic acid catabolic process
antibiotic metabolic process
negative regulation of apoptotic signaling pathway
0 5 10 15
Count
Wnk1
Pgam2
Clu
Tmem52b
Calb1
Slc12a3
Emx1
S100g
Kl
Defb1
Scd2
Sgms2
Tmem213
Ier3
Cd9
Sh3bgrl3
Lhx1os
Klk1
Vdr
Lhx1
0.6 0.7 0.8
cor
d
Mouse kidney distal tubule K2 module
p.adjust
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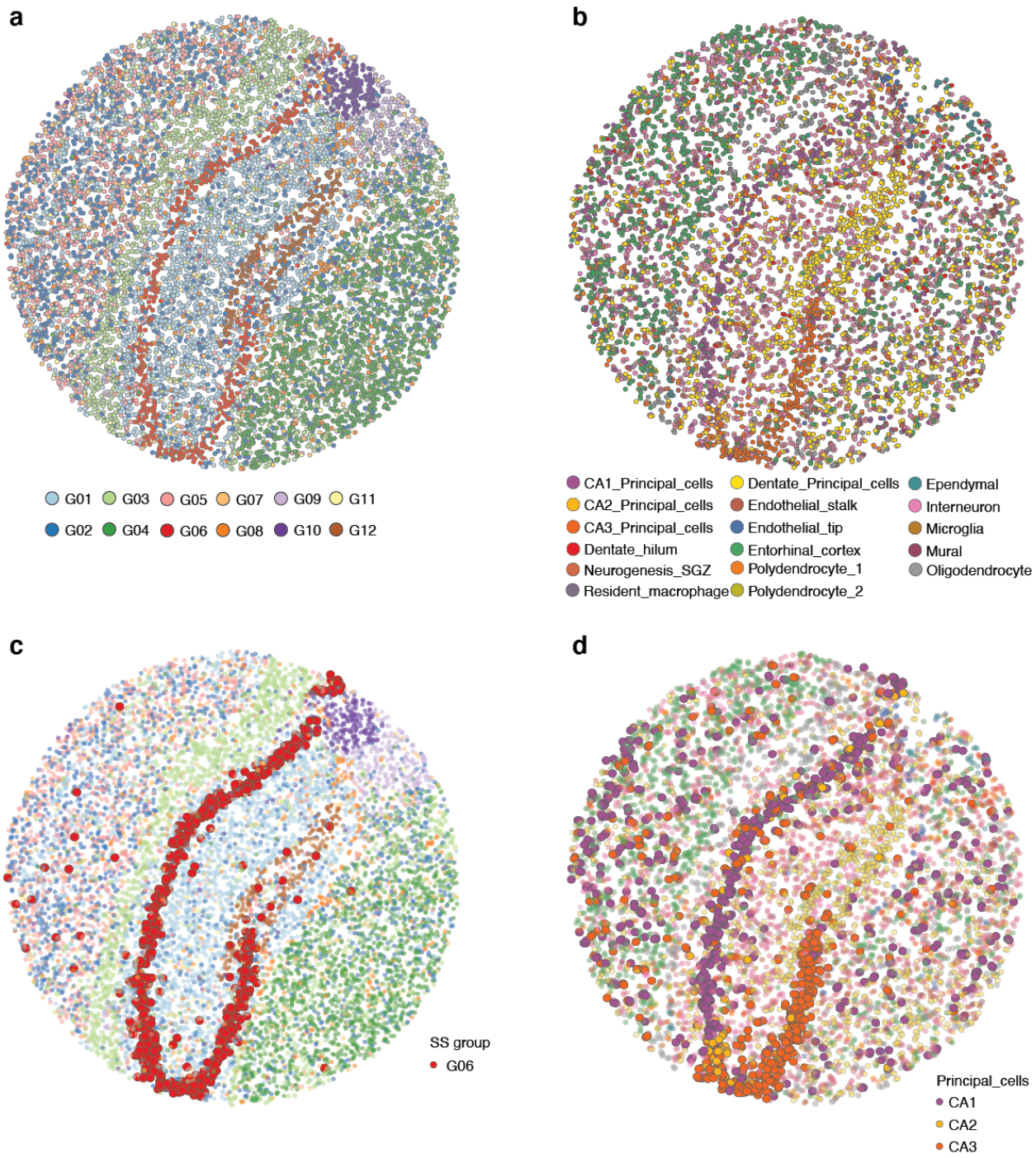
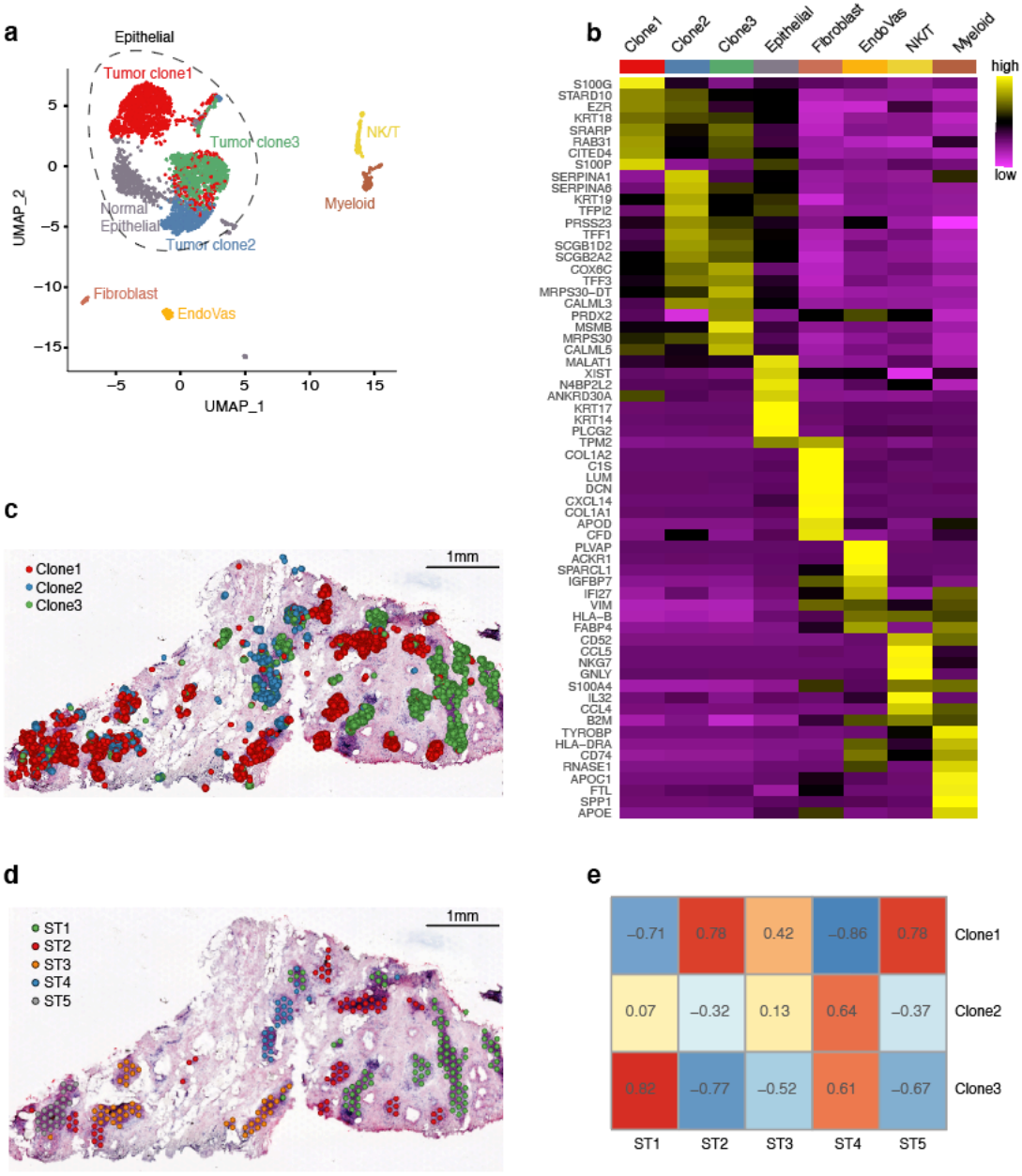
p_val_adj < 0.01
mitochondrion organization
cell−matrix adhesion
generation of precursor metabolites and energy
cell−substrate junction assembly
ATP metabolic process
purine ribonucleotide metabolic process
purine−containing compound metabolic process
purine ribonucleoside monophosphate metabolic process
purine ribonucleoside triphosphate metabolic process
regulation of response to wounding
ribonucleotide metabolic process
purine nucleoside monophosphate metabolic process
regulation of epithelial cell migration
negative regulation of response to endoplasmic reticulum stress
ribonucleoside monophosphate metabolic process
0 5 10 15
Count
Slc12a1
Nccrp1
Cldn10
Wfdc15b
Ppp1r1b
Slc5a3
Ptger3
Gpx6
Umod
Ckb
Cldn19
Enpp2
Tmem207
Mrps6
Clcnka
Gm17750
Epcam
Car2
Cd24a
Pla1a
0.5 0.6 0.7
cor
e
Raw interactions
Prin
Macro
ProxTub
Inter
RenaCorp
DistTub
T
VSMC
TLLH
Vasc
f
Spatial constrained interactions
Prin
Macro
ProxTub
Inter
RenaCorp
DistTub
T
VSMC
TLLH
Vasc
g
Selected LR pairs
Flt1
FLT1_KDR
Cxcr6
Kdr
Ackr3
Vegfa
Vegfa
Mif
Cxcl16
DisTub
Prin
RenaCorp
Macro
ProxTub
T
Vasc

Supplementary Figure 5 – Mouse brain and kidney co-expression module enrichment analysis and mouse kidney spatial constrained cell-cell interaction analysis

(**a-b**) GO enrichment analyses (left) and module-correlated genes (right) for mouse brain L5 IT K1 and K2 modules, respectively. (**c-d**) GO enrichment analyses (left) and module-correlated genes (right) for mouse kidney distal tubule K1 and K2 modules, respectively. (**e**) Summary of cell-cell interaction network of mouse kidney cells using CellChat. (**f**) Spatial constrained cell-cell interaction network of mouse kidney cells. (**g**) Chord plot showing some identified ligand-receptor pairs. The edge width in (**e**) and (**f**) represent the total interaction strength; and the arc width in (**g**) represent the interaction score calculated from CellChat. The direction of the arrows goes from the ligands to the receptors. In **a**-**d**, *P* values were calculated by hypergeometric distribution and adjusted by Benjamini-Hochberg method. Module-correlated genes were determined by the CellTrek *FindCorMarkers* module. Spearman correlation was performed and *P* values were adjusted by the Bonferroni method.

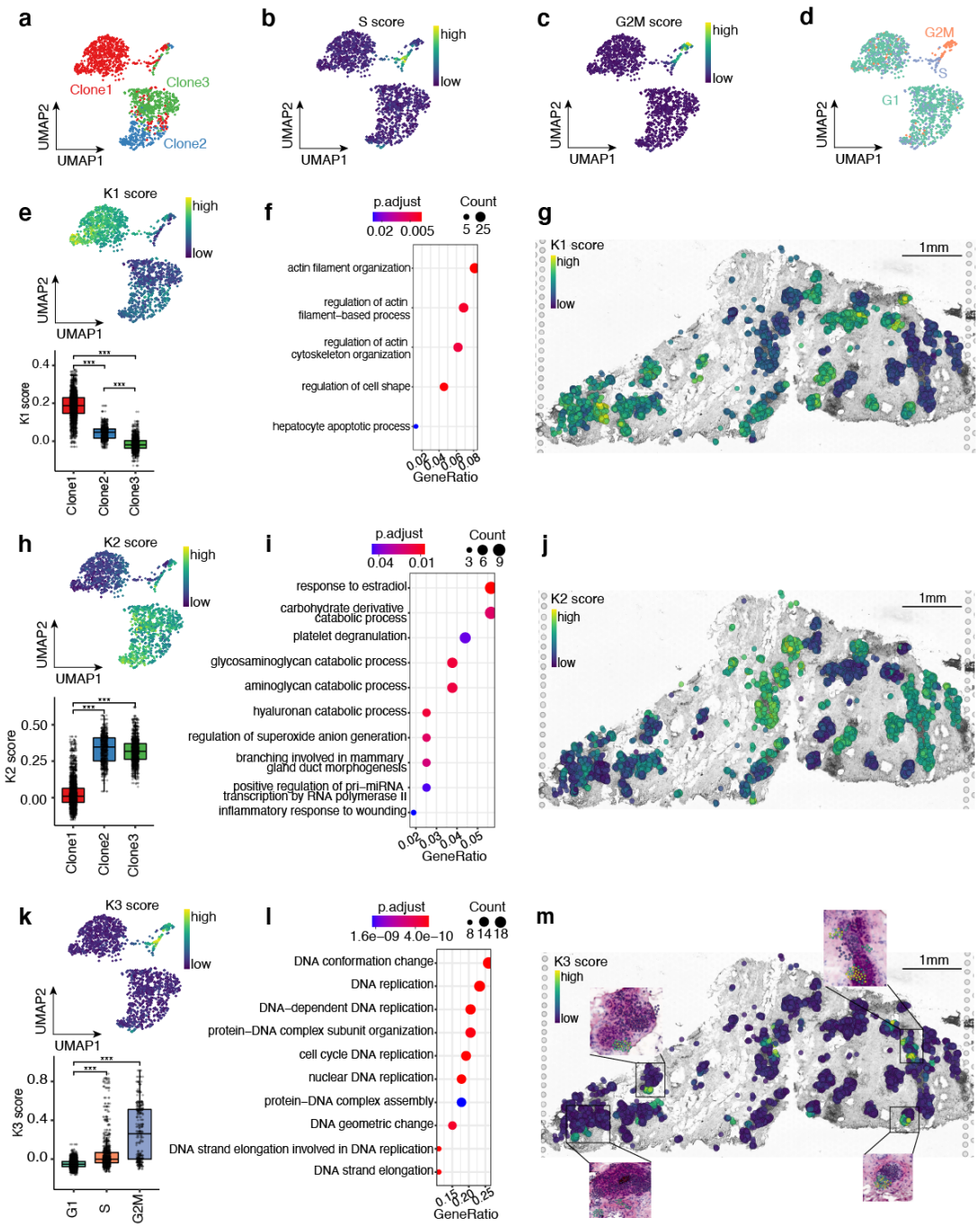


Supplementary Figure 6 – CellTrek analysis of mouse hippocampus Slide-seq data
(**a**) Unsupervised clustering of mouse hippocampus Slide-seq data identified 12 clusters. (**b**) CellTrek spatial mapping of scRNA-seq data to the tissue section. (**c**) Slide-seq data with G06 group highlighted. (**d**) CellTrek map with three principal cells (CA1, CA2 and CA3) highlighted which corresponded to G06.



Supplementary Figure 7 – ScRNA-seq and ST analyses of DCIS1

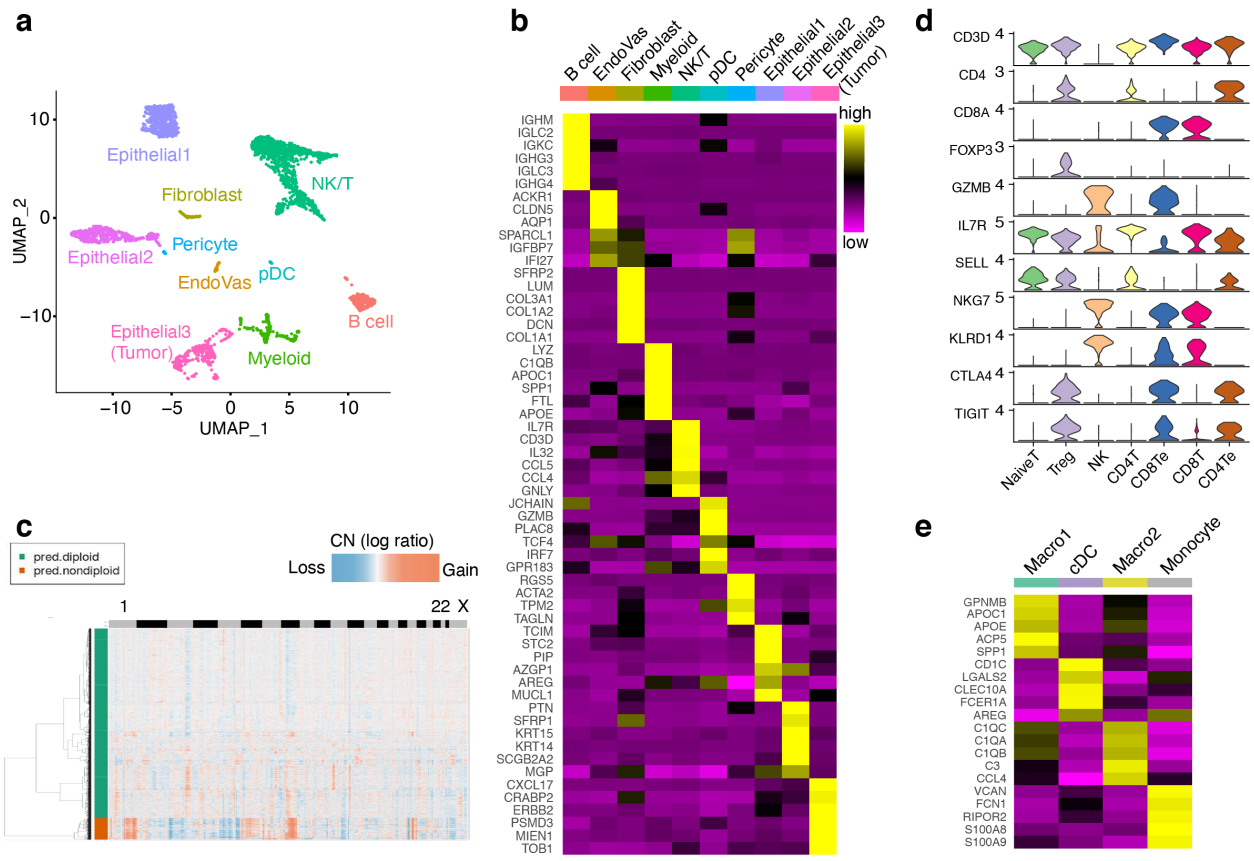
(**a**) UMAP of DCIS1 scRNA-seq data. (**b**) Consensus heatmap of scRNA-seq data for top differentially expressed genes for each cell type. (**c**) CellTrek result of three tumor subclones (clone1-3) which is also presented in **Fig. 4e**. (**d**) Unsupervised clustering of ST tumor spots identified five clusters (ST1-5). (**e**) Gene expression-based Spearman correlation analysis between three tumor subclones in the scRNA-seq data (clone1-3) and five ST clusters (ST1-5).



Supplementary Figure 8 – Spatial co-expression patterns of tumor cells in DCIS1

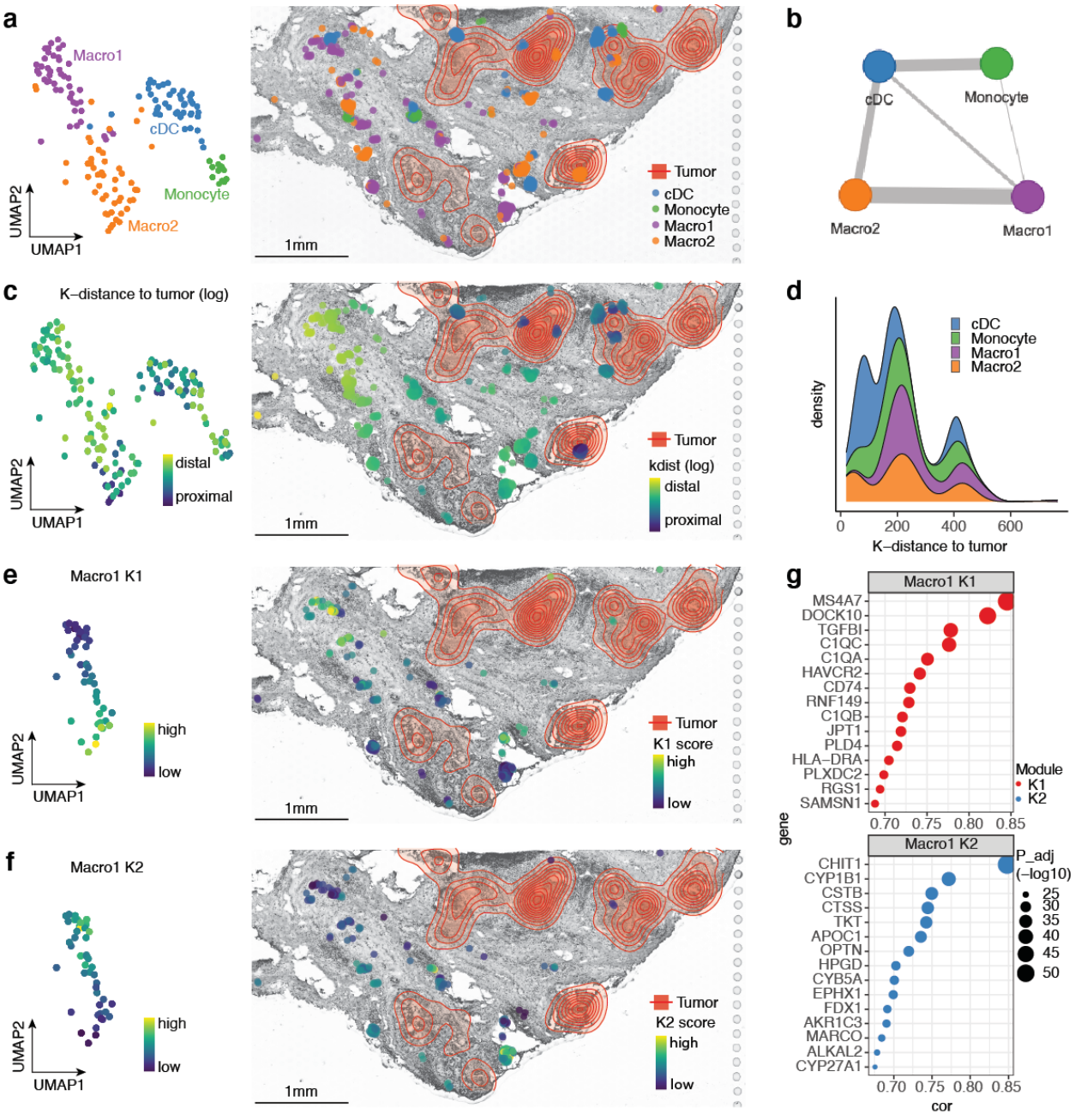
(**a**) Transcriptome-based UMAP of three tumor subclones identified from DCIS1 scRNA-seq data (Clone1 n = 1,295; Clone2 n = 523; Clone3 n = 862). Seurat-based cell cycle analysis on tumor cells showed S scores (**b**), G2M scores (**c**) and categorical cell cycle phases (G1 n = 2,060; G2M n = 508; S n = 1,019) (**d**). (**e**) K1 module scores on the UMAP (upper) and boxplot across tumor subclones (lower). (**f**) GO enrichment analysis for the K1 module. (**g**) CellTrek map of K1 module scores. (**h**) K2 module scores on the UMAP (upper) and boxplot across tumor subclones (lower). (**i**) GO enrichment analysis for the K2 module. (**j**) CellTrek map of K2 module scores. (**k**) K3 module scores on the UMAP (upper) and boxplot across different cell cycle phases (lower). (**l**) GO enrichment analysis for the K3 module. (**m**) CellTrek map of K3

module scores. In e, h, and k two-sided Wilcoxon rank-sum tests were performed. Boxplots show the median with interquartile ranges (25–75%); whiskers extend to 1.5X the interquartile range from the box. In **f**, **i** and **l**, *P* values were calculated by hypergeometric distribution and further adjusted by Benjamini-Hochberg method.



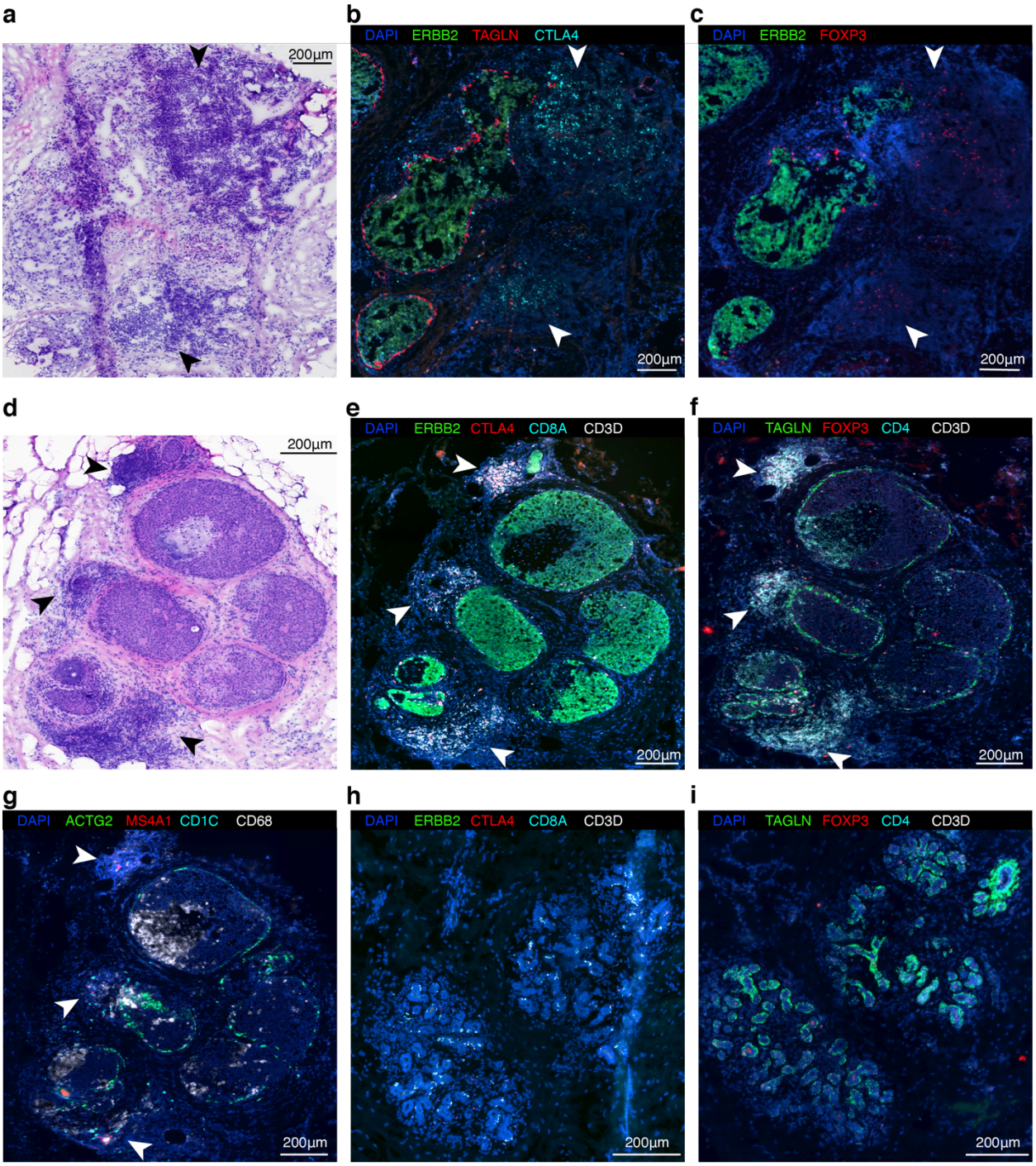
Supplementary Figure 9 – ScRNA-seq data analysis of DCIS2

(**a**) UMAP of DCIS2 scRNA-seq data, with clusters corresponding to different cell types. (**b**) Consensus heatmap of top differentially expressed genes for each cell cluster. (**c**) Heatmap of CopyKAT inferred copy number profiles from the scRNA-seq data and hierarchical clustering of cells. (**d**) Violin plots showing expression of canonical T cell state markers across different T cell states. (**e**) Consensus heatmap showing expression of top differentially expressed genes across different myeloid cell states.



Supplementary Figure 10 – CellTrek analyses of myeloid cell states in DCIS2

(**a**) UMAP representation of myeloid cell states (left) and corresponding CellTrek map (right) (**b**) Spatial graph of myeloid cell states colocalization using SColoc. (**c**) UMAP (left) and CellTrek map (right) of myeloid cells K-distance to tumor cells. (**d**) Density plot of K-distances to tumor cells for each myeloid cell state. (**e**) UMAP (left) and CellTrek map (right) of Macro1 K1 module scores. (**f**) UMAP (left) and CellTrek map (right) of Macro1 K2 module scores. (**g**) Macro1 K1 (top) and K2 (bottom) module top correlated genes. The contour plots represent the densities of tumor cells. Module-correlated genes were determined by the CellTrek *FindCorMarkers* module. Spearman correlation was performed and *P* values were adjusted by the Bonferroni method.


a
200μm
b
DAPI ERBB2 TAGLN CTLA4
200μm
c
DAPI ERBB2 FOXP3
200μm
d
200μm
e
DAPI ERBB2 CTLA4 CD8A CD3D
200μm
f
DAPI TAGLN FOXP3 CD4 CD3D
200μm
g
DAPI ACTG2 MS4A1 CD1C CD68
200μm
h
DAPI ERBB2 CTLA4 CD8A CD3D
200μm
i
DAPI TAGLN FOXP3 CD4 CD3D
200μm

Supplementary Figure 11 – Orthogonal validation of tumor and immune cell spatial distributions using RNAscope assay in DCIS2 and DCIS3

(a) H&E image of the DCIS2 tumor region. (b) RNAscope assay for *ERBB2, TAGLN* and *CTLA4* in DCIS2 tumor region. (c) RNAscope assay for *ERBB2* and *FOXP3* in DCIS2 tumor region. Data in (a-c) were collected from adjacent tissue slides. (d) H&E image of the DCIS3 tumor region. (e) RNAscope assay for *ERBB2*, *CTLA4, CD8A* and *CD3D* in DCIS3 tumor region. (f) RNAscope assay for *TAGLN*, *FOXP3*, *CD4* and *CD3D* in DCIS3 tumor region. (g) RNAscope assay for *ACTG2*, *MS4A1*, *CD1C* and *CD68* in DCIS3 tumor region. (h) RNAscope assay for *ERBB2*, *CTLA4*, *CD8A* and *CD3D* in DCIS3 adjacent normal lobular region. (i) RNAscope assay for *TAGLN*, *FOXP3*, *CD4* and *CD3D* in DCIS3 adjacent normal lobular region. Arrows point to different immune cell markers near the DCIS tumor ducts.

Supplementary Tables

Supplementary Table 1 Parameter settings for different conditions in simulation-1

	scRNA-seq				ST			
Condition	Counts	Cell Proportion	Median Count	Median Gene	Counts	Spot Proportion	Median Count	Median Gene
1	10	0.2	2815	494	50	0.2	15732	1526
2	10	0.4	2328	414	50	0.4	12516	1356
3	15	0.4	2195	414	75	0.4	12489	1356
4	15	0.5	1781	207	75	0.5	7006	338
5	20	0.5	1622	192.5	100	0.5	5055	320
6	20	0.6	929	63	100	0.6	2257	39
7	25	0.6	671	42	125	0.6	1537	23
8	25	0.7	444.5	31	125	0.7	1183	19

Counts means the UMI counts for each gene we downsampled and Cell/Spot Proportion means the proportion of cells/ST spots were used for downsampling. Median Count and Gene mean the median of UMI count and gene number after downsampling.

Supplementary Table 2 Parameter settings for different conditions in simulation-2

Condition	Spatial Deviation
1	30
2	60
3	90
4	120
5	150
6	180
7	210
8	240

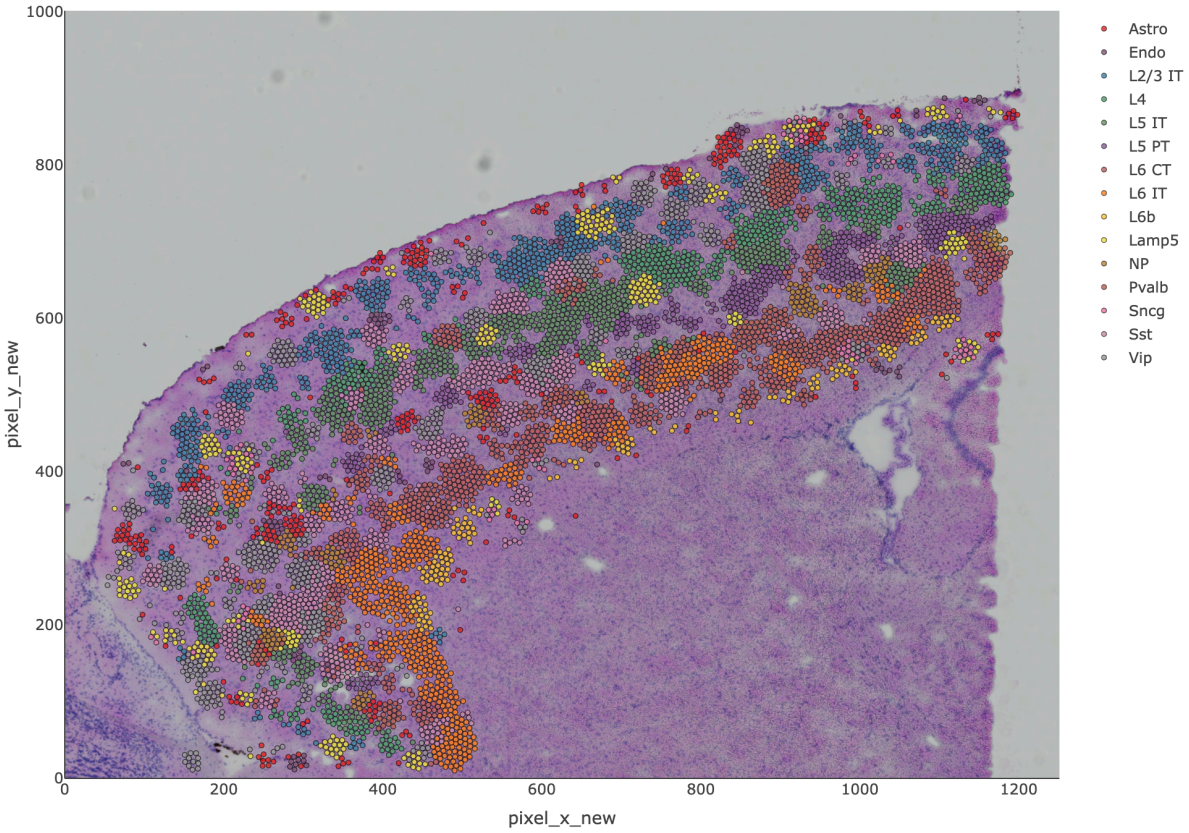
Spatial Deviation means the deviation of the spatial gaussian noise we added to the cell.

Supplementary Table 3 Parameter settings for different marker points in simulation-3

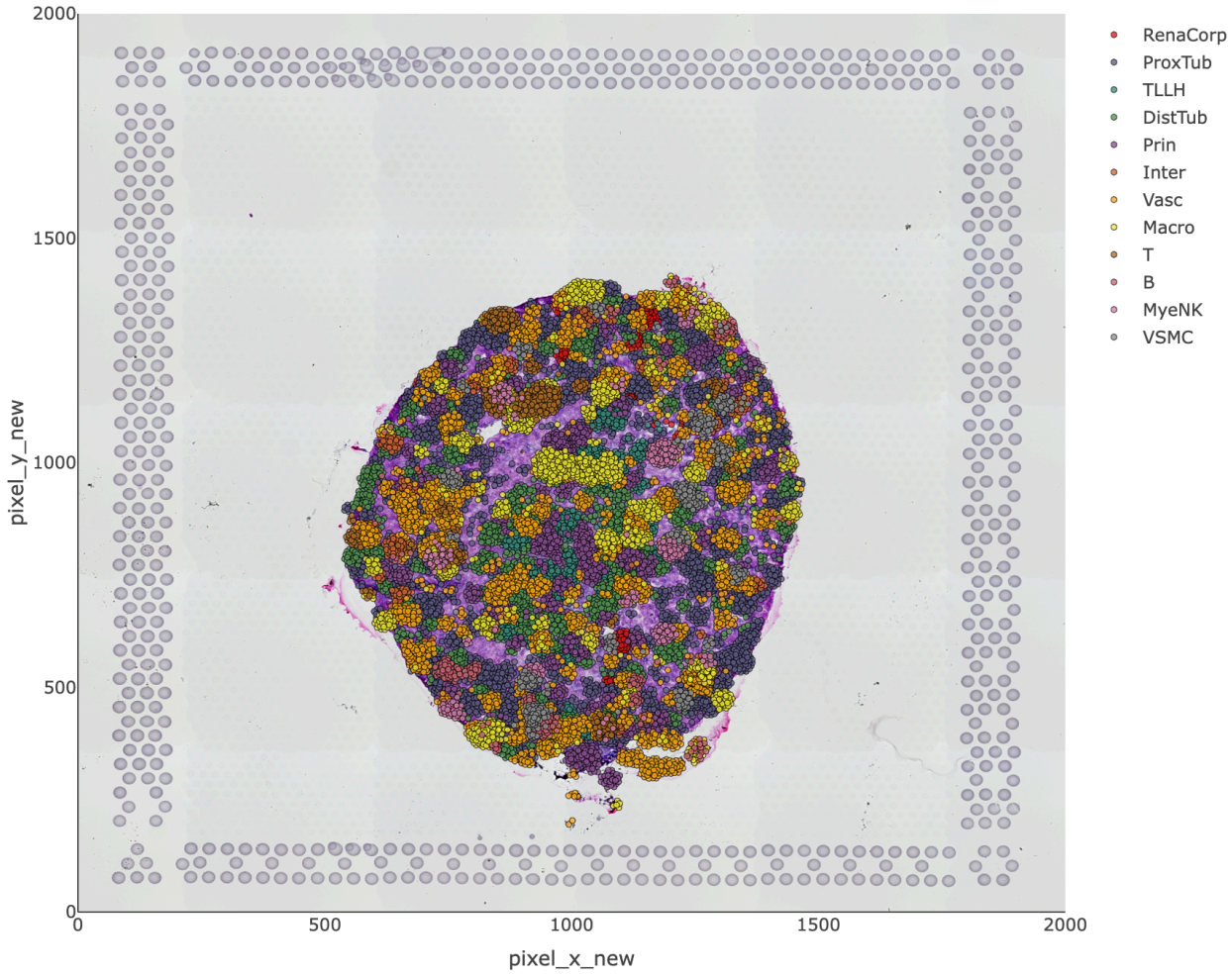
MarkerPoint	X-coordinate	Y-coordinate
1	350	300
2	1300	1100
3	1950	800
4	1500	600
5	600	800
6	1000	1000
7	900	400
8	1700	1100

Supplementary Data

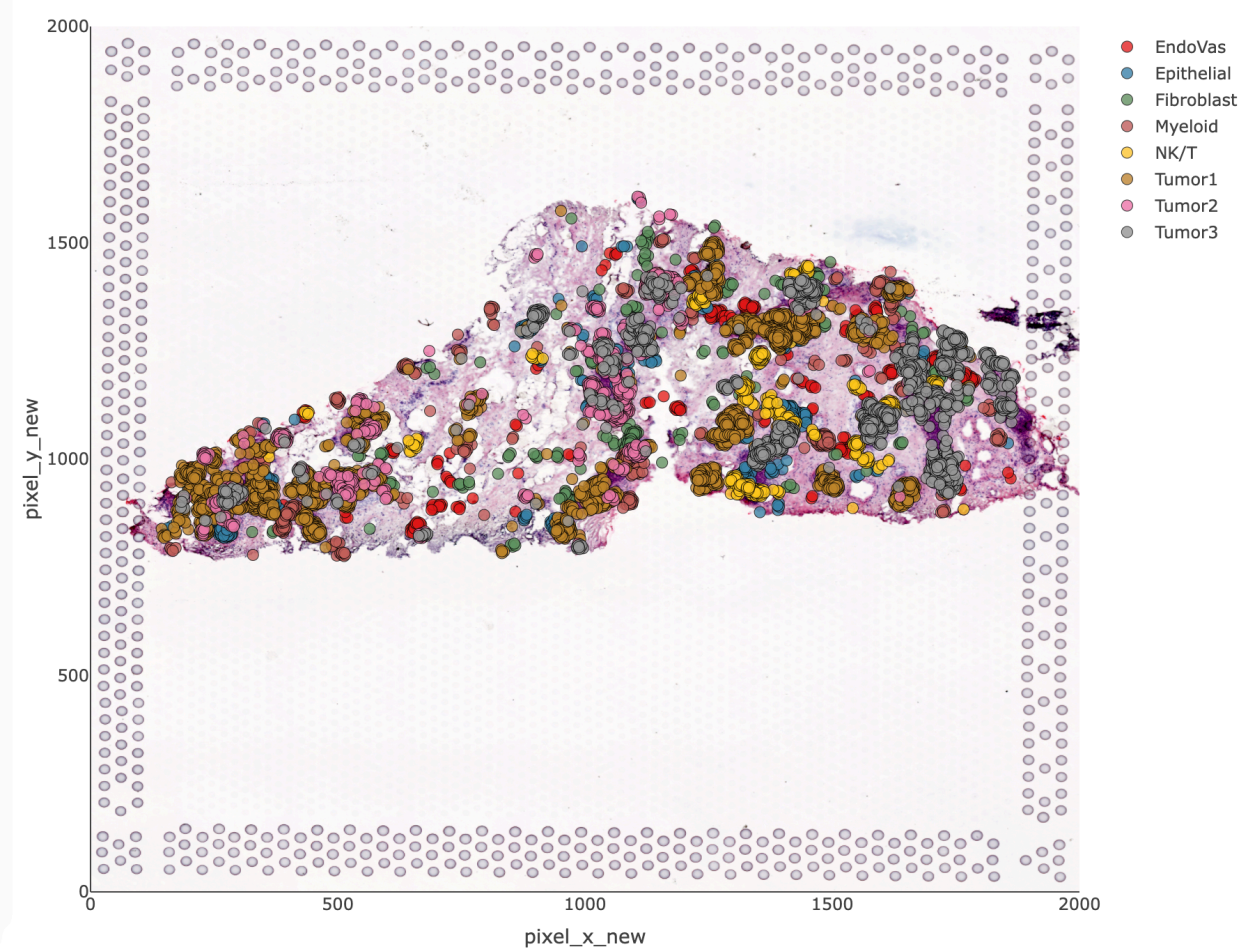
Supplementary Data 1 – Screenshot of the CellTrek result for mouse brain data



Supplementary Data 2 – Screenshot of the CellTrek result for mouse kidney data



Supplementary Data 3 – Screenshot of the CellTrek result for DCIS1 data



Supplementary Data 4 – Screenshot of the CellTrek result for DCIS2 data